Rethinking Hard Thresholding Pursuit: Full Adaptation and Sharp Estimation

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Abstract—**Hard Thresholding Pursuit (HTP) has aroused increasing attention for its robust theoretical guarantees and impressive numerical performance in non-convex optimization. This paper consider a high-dimensional linear regression model with n observations, p predictors, and an unknown s^*-sparse signal $\beta^* \in \mathbb{R}^p$ corrupted by noise of magnitude σ. We introduce a novel tuning-free procedure, namely Full-Adaptive HTP (FAHTP), that simultaneously adapts to both the unknown sparsity and signal strength of the underlying model. Our theoretical analysis rigorously characterizes the iterative thresholding dynamics of FAHTP, offering refined theoretical insights. In specific, under the beta-min condition $\min_{\{i:\beta_i^*\neq 0\}} |\beta_i^*| \geq C\sigma(\log p/n)^{1/2}$, FAHTP achieves oracle estimation rate $\sigma(s^*/n)^{1/2}$, highlighting its theoretical superiority over convex competitors such as LASSO and SLOPE, and recovers the true support set exactly. More importantly, even without the beta-min condition, FAHTP achieves a tighter error bound than the classical minimax rate with high probability. The comprehensive numerical experiments substantiate our theoretical findings, underscoring the effectiveness and robustness of the proposed FAHTP.**

Index Terms—**Exact support recovery, full adaptation, hard thresholding pursuit, minimax optimality, oracle estimation.**

I. Introduction

Consider a linear model

$$y = X\beta^* + \xi,$$

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where $y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n\times p}$ and $\beta^* \in \mathbb{R}^p$ are the response vector, the design matrix, and the underlying regression coefficient, respectively. And $\xi \in \mathbb{R}^n$ is the sub-Gaussian random error with scale parameter σ^2. In the high-dimensional learning framework, we focus on the case where $p \gg n$ and the coefficient β^* is s^*-sparse, and we define $S^* := \{i \in [p] : \beta_i^* \neq 0\}$ as the true support set of β^*.

One typical learning approach under empirical risk minimization is the best-subset selection problem, formulated as

$$\min_{\beta\in\mathbb{R}^p} \mathcal{L}_n(\beta), \quad s.t. \ \|\beta\|_0 \leq s, \tag{1}$$

where $\mathcal{L}_n(\beta) = \|y - X\beta\|_2^2/(2n)$ and s is some given positive integer controlling the sparsity level. However, problem (1) is an NP-hard problem [1], indicating it lacks a polynomial-time solution.

Among the numerous solutions that aim at approximately solving the ℓ_0-sparse problem (1), the Iterative Hard Thresholding (IHT, [2]) algorithm stands out as an extensively studied iterative greedy selection approach. It keeps the s largest values of the gradient descent in each step via a hard thresholding procedure, yielding an estimator with s nonzero components. To enhance empirical performance, [3] introduced a debiasing step at each iteration, employing an ordinary least-squares (OLS) estimator—a method they named Hard Thresholding Pursuit (HTP). The extensive literature on HTP and its variants includes both statistics and optimization [4], [5], [6]. In practice, HTP-style methods have found their wide applications in deep neural networks pruning [7], genetic data analysis [8], [9], and Alzheimer's Disease Neuroimaging Initiative study [10]. In theory, [11] relaxed the bounded restriction on the RIP-type condition for $s \gg s^*$, and showed the minimax optimality for parameter estimation of IHT up to a logarithm factor. Reference [4] analyzed the parameter estimation and support recovery of HTP for both $s = s^*$ and $s \gg s^*$. Additionally, [5] established the ℓ_2 and ℓ_∞ estimation error bounds for $s \geq s^*$.

A. Main Results

Our work focuses on the estimation and minimax adaptation properties of HTP. In contrast to techniques based on complexity theory and empirical processes used in convex estimators like LASSO [12], or non-convex algorithms such as SCAD and MCP [13], [14], our approach offers a detailed analysis of the iterative thresholding dynamics, providing a deeper understanding of the intrinsic properties of the hard thresholding estimator. By relying solely on the RIP assumption (2), we

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demonstrate that the HTP algorithm has the following key properties:

Proposition 1: We divide S^* into

$$S_1^* := \{i \in S^* : |\boldsymbol{\beta}_i^*| \geq C_1\sigma(\log p/n)^{1/2}\}, \quad S_2^* := S^* \setminus S_1^*.$$

Assume $|S_1^*|\log p/n \to 0$, $|S_1^*|/p \to 0$ and $|S_1^*| \to \infty$. Let s be the input parameter in Algorithm 1. Given $s = |S_1^*|$, with probability tending to 1, the estimator of HTP has

$$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2 \leq C_2\sigma\left(\frac{|S_1^*|}{n}\right)^{1/2} + C_3\|\boldsymbol{\beta}^*_{S_2^*}\|_2. \tag{2}$$

Proposition 1 shows that HTP can achieve a tighter estimation upper bound better than the classical minimax rate $\sigma(s^*\log p/n)^{1/2}$. Specifically, HTP successfully detects the large signal set S_1^* and provides unbiased estimates for these signals, while the hard thresholding operator simultaneously shrinks the small signals in S_2^*. This combination of advantages allows HTP to surpass the classical minimax rate and derive a sharper error bound (2).

Based on the estimation results in Proposition 1, we propose a novel tuning-free strategy (FAHTP, Algorithm 2) to select the sparsity parameter s.

Proposition 2: Assume that $\min_{i\in S^*}|\boldsymbol{\beta}_i^*| \geq C_1\sigma(\log p/n)^{1/2}$, and $s^*\log p/n \to 0$, $s^*/p \to 0$, $s^* \to \infty$. Then, by the proposed FAHTP, with probability tending to 1, the estimator of HTP has

$$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2 \leq C_2\sigma\left(\frac{s^*}{n}\right)^{1/2}$$

and $\hat{\boldsymbol{\beta}}$ can recover the true support set S^* exactly.

Proposition 2 demonstrates that FAHTP achieves exact support recovery under the optimal beta-min condition, as established in [15]. These results indicate that FAHTP successfully inherits the strengths of the hard thresholding operator in the context of the Gaussian sequence model [15], [16], under the RIP assumption.

B. Related Literature and Comparison

Over the past few decades, convex relaxation algorithms to the problem (1), such as LASSO [17] and SLOPE [18], have been widely studied. While these convex algorithms can achieve the minimax optimal rate $\sigma\{s^*\log(p/s^*)/n\}^{1/2}$ [12], [19], they may not be admissible in certain circumstances. In the Gaussian sequential model, [20] uncovered a phase transition phenomenon in the minimax rates under the beta-min condition $\min_{i\in S^*}|\boldsymbol{\beta}_i^*| \geq C\sigma\{\log(p/s^*)/n\}^{1/2}$. They demonstrated that, under this condition, the minimax rate improves to the oracle estimation rate $\sigma(s^*/n)^{1/2}$, independent of the dimension p. This finding suggests that an appropriate beta-min condition can lead to sharper estimation error bounds. However, [21] showed that convex algorithms can only achieve the original minimax optimal rate when the beta-min condition holds.

The above observations lead to an important question: when and why does the HTP algorithm outperform the convex counterparts? To be specific, we aim to investigate when the HTP can exceed the minimax rate and achieve the oracle estimation rate $\sigma(s^*/n)^{1/2}$.

References [6] and [22] investigated the theoretical guarantees of the HTP-style method under a stronger beta-min condition. Specifically, [6] proposed a tuning-free procedure using a splicing technique and demonstrated that it could recover the support set S^* exactly under a stronger condition $\min_{i\in S^*}|\boldsymbol{\beta}_i^*| \geq C\sigma\{s^*\log p\log\log n/n\}^{1/2}$. Reference [22] established the fast rates for the empirical risk of IHT relying on condition $\min_{i\in S^*}|\boldsymbol{\beta}_i^*| \geq C\sigma\{s^*\log p/n\}^{1/2}$. Recent work [23], [24] proved the variable selection and FDR-control behavior of IHT-style methods based on condition $\min_{i\in S^*}|\boldsymbol{\beta}_i^*| \geq C\sigma(\log p/n)^{1/2}$. These studies also assume $s^* = O(\log p)$, a condition not required in our analysis. Notably, the aforementioned works did not establish the oracle estimation rate for HTP-style methods.

To our knowledge, the minimax adaptation of HTP-style methods to unknown noise level σ and sparsity s^* still remains unexplored. Reference [25] conducted Gaussian model selection based on the penalization of complexity with a known noise level σ. Building on this approach, [26] proposed a novel information criterion that is minimax adaptive to both unknown s^* and σ. However, this procedure relies on an exhaustive search with subset sizes less than $n/4$, making it inefficient for practical implementation. Therefore, developing a minimax adaptive procedure using efficient methods such as HTP is a significant direction for high-dimensional model selection.

The classical theory of LASSO [12] requires prior knowledge of noise level σ and true sparsity s^* to achieve sharp optimality, with the regularization parameter set as $\lambda = C\sigma\{\log(p/s^*)/n\}^{1/2}$. To address these limitations, significant progress has been made in developing adaptive variants of LASSO-type estimators. The Square-Root LASSO [27] introduced a σ-free estimator through minimization of the ℓ_2-norm of residuals, while the SLOPE estimator [18], [28] attains minimax optimal estimation rates without knowledge of s^* through sorted ℓ_1 penalties. Subsequent work [29] established minimax adaptivity to both σ and s^* for Square-Root LASSO and Square-Root SLOPE. Further developments [12] show that LASSO with Lepski's method achieves sharp adaptation to s^*. Reference [30] proposed a two-stage Square-Root SLOPE estimator that achieves exact support recovery without knowledge of σ or s^*. Our work on parameter selection for HTP attains simultaneous adaptation to σ and s^*. Furthermore, it adapts to the beta-min condition, enabling the attainment of the oracle rate, whereas convex methods achieve only minimax optimal rates.

Thus, the main drawbacks in the existing analysis for HTP are listed as follows:

- Firstly, the signal condition required in [6] and [22] is relatively stronger than the condition (see in Theorem 3) listed in this paper, which is the optimal beta-min condition proved by [20].
- Secondly, whether the classical HTP-style algorithm can achieve the oracle estimation rate $\sigma(s^*/n)^{1/2}$ remains unclear, indicating further analysis is needed.

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- Lastly, previous studies about HTP (e.g., [4], [5]) do not establish minimax adaptivity to both unknown noise level σ and sparsity s^*.

Some of the works mentioned below theoretically inspire us. Reference [15] analyzed variable selection from the non-asymptotic perspective with hamming loss, and this can be linked to estimation risk [20]. In recent work, [31] introduced a novel IHT-style procedure, extending the theory from [15] and [20] to linear regression. This procedure involves dynamically updating the threshold geometrically at each step until it reaches a universal statistical threshold. Our procedure, choosing the sparsity s exactly instead of dynamically updating the threshold, offers one main practical advantage as follows. Because the model size is discrete while the tuning parameter is continuous, the tuning for model size s is more intuitive in practice and is computationally more efficient compared to the M-estimators.

C. Main Contributions

We summarize the main contributions of our paper as the following two points:

- Under the beta-min condition $\min_{i\in S^*}|\boldsymbol{\beta}_i^*| \geq C\sigma\{\log(p/s^*)/n\}^{1/2}$, our study reveals that the HTP algorithm surpasses the minimax rate $\sigma\{s^*\log(p/s^*)/n\}^{1/2}$ and attains the oracle estimation rate $\sigma(s^*/n)^{1/2}$. Moreover, we characterize the dynamic signal detection properties of the HTP algorithm. Because the hard thresholding performs signal detection through iterations, the algorithm can shrink signals according to magnitudes, which leads to the adaptation to signals as well as the oracle estimation rate.
- We propose a novel tuning-free procedure (FAHTP, Algorithm 2) to simultaneously adapt to the unknown sparsity s^*, noise level σ, and signal strength. Theorem 2 guarantees that the FAHTP procedure achieves minimax optimality with unknown s^* and σ in the first stage. Additionally, under beta-min condition $\min_{i\in S^*}|\boldsymbol{\beta}_i^*| \geq C\sigma(\log p/n)^{1/2}$, Corollary 4 guarantees that FAHTP further attains the oracle estimation rate and exactly recovers the support set S^* in the second stage.
  Moreover, when the beta-min condition is not satisfied, FAHTP can still derive a tighter error bound than the minimax rate.

Based on the above two points, solely under RIP assumption, our proposed algorithm is minimax adaptive to the signal strength and support size s^*, which fully matches the classical minimax rate and oracle rate [20].

D. Organization

The present paper is organized as follows. In Section II, we establish the estimation error bounds of HTP when $s \geq s^*$, which match the known minimax lower bounds and demonstrate the optimality of HTP. Based on these results, we propose a minimax adaptive procedure with unknown s^* and σ. In Section III, under the beta-min condition, we prove that the HTP estimator can attain the oracle estimation rate and achieve the exact recovery of the true support set when $s = s^*$. In Section IV, we extend the analysis to general signals, i.e., without the beta-min conditions, and show that our estimator can obtain tighter error bounds with high probability. Section V presents numerical experiments that illustrate our theoretical findings. Lastly, our conclusions are drawn in Section VI. Detailed proofs of the theory and additional simulations are available in the appendix. More importantly, we provide a road map of the proof of the main results in the appendix.

E. Notations

For the given sequences a_n and b_n, we say that $a_n = O(b_n)$ (resp. $a_n = \Omega(b_n)$) when $a_n \leq cb_n$ (resp. $a_n \geq cb_n$) for some positive constant c. We write $a_n \asymp b_n$ if $a_n = O(b_n)$ and $a_n = \Omega(b_n)$ while $a_n = o(b_n)$ corresponds to $a_n/b_n \to 0$ as n goes to infinity. $\log_a(\cdot)$ denotes the logarithm with base a and $\log(\cdot)$ denotes the natural logarithm (i.e., with base e). Denote $\lceil\cdot\rceil$ as the ceiling function, Define $[m]$ as the set $\{1, 2, \ldots, m\}$, and $\mathrm{I}(\cdot)$ as the indicator function. Let $x \vee y$ be the maximum of x and y, while $x \wedge y$ is the minimum of x and y. Denote $S^* = \{i : \boldsymbol{\beta}_i^* \neq 0\} \subseteq [p]$ as the support set of $\boldsymbol{\beta}^*$. For any set S with cardinality $|S|$, let $\boldsymbol{\beta}_S = (\boldsymbol{\beta}_j, j \in S) \in \mathbb{R}^{|S|}$ and $\boldsymbol{X}_S = (\boldsymbol{X}_j, j \in S) \in \mathbb{R}^{n\times|S|}$, and let $(\boldsymbol{X}^\top\boldsymbol{X})_{SS} \in \mathbb{R}^{|S|\times|S|}$ be the submatrix of $\boldsymbol{X}^\top\boldsymbol{X}$ whose rows and columns are both listed in S. For a vector $\boldsymbol{\beta}$, denote $\|\boldsymbol{\beta}\|_2$ as its Euclidean norm and $\|\boldsymbol{\beta}\|_0$ as the count of the nonzero elements of $\boldsymbol{\beta}$. Let $\mathrm{supp}(\boldsymbol{\beta})$ be the support set of $\boldsymbol{\beta}$. For a matrix $\boldsymbol{X}$, denote $\|\boldsymbol{X}\|_2$ as its spectral norm and $\|\boldsymbol{X}\|_F$ as its Frobenius norm. Denote $\mathbb{I}_p$ as the $p\times p$ identity matrix. For two sets A and B, define the symmetric difference of A and B as $A\Delta B = (A\backslash B)\cup(B\backslash A)$. Let $C, C_0, C_1, \ldots$ denote positive constants whose actual values vary from time to time. To facilitate computation, we assume $\|\boldsymbol{X}_j\|_2 = n^{1/2}$ for all $j \in [p]$.

Notably, the asymptotic results are considered as $p \to \infty$ when all other parameters of the problem, i.e., n, s^*, depend on p in such a way that $n = n(p) \to \infty$. For brevity, the dependence of these parameters on p will be further omitted in the notation.

II. Minimax Optimality and Adaptation of HTP

In this section, we present the nonasymptotic error bounds for the HTP estimator. We first provide the HTP algorithm in Section II-A. Then, the ℓ_2 error bounds are established in Section II-B. Based on these results, Section II-C develops a minimax adaptive procedure adaptive to the unknown sparsity and noise level.

A. HTP Algorithm

To enforce the sparsity of the solution, HTP applies a hard thresholding operator $\mathcal{T}_s(\cdot)$ to the gradient descent in each iteration, where the operator $\mathcal{T}_s(\boldsymbol{\beta})$ is a non-linear operator that sets all but the largest (in absolute sense) s elements of $\boldsymbol{\beta}$ to zero. Then, it incorporates a debiasing step into each step. The HTP algorithm is presented in Algorithm 1.

We can decompose the iterative term into the following parts:

$$\boldsymbol{\beta}^t + \boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}^t)/n = \boldsymbol{\beta}^* + \underbrace{\boldsymbol{\Phi}(\boldsymbol{\beta}^* - \boldsymbol{\beta}^t)}_{\text{optimization error}} + \underbrace{\boldsymbol{\Xi}}_{\text{statistical error}},$$

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Algorithm 1 Hard Thresholding Pursuit (HTP) Algorithm

Require: $\boldsymbol{X}$, $\boldsymbol{y}$ and s.

Initialize $t = 0$, $\boldsymbol{\beta}^t = \mathbf{0}$ and $S^t = \emptyset$.
while $S^{t+1} \neq S^t$, **do**
  $\tilde{\boldsymbol{\beta}}^{t+1} = \mathcal{T}_s\left(\boldsymbol{\beta}^t + \boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}^t)/n\right)$.
  $S^{t+1} = \mathrm{supp}(\tilde{\boldsymbol{\beta}}^{t+1})$.
  $\boldsymbol{\beta}^{t+1} = \arg\min\{\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2,\ \mathrm{supp}(\boldsymbol{\beta}) = S^{t+1}\}$.
  $t = t + 1$.
end while

Ensure: $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta}^t$.

where $\boldsymbol{\Phi} := \boldsymbol{X}^\top\boldsymbol{X}/n - \mathbb{I}_p$ and $\boldsymbol{\Xi} := \boldsymbol{X}^\top\boldsymbol{\xi}/n$. From the decomposition, we can have a view that the error at each step mainly comes from two parts, that is, the estimation error incurred by $\boldsymbol{\beta}^t$ and the statistical error incurred by the white noise $\boldsymbol{\xi}$. In the subsequent section, we discuss how to control these two sources of errors during iterations.

B. ℓ_2- Error Bounds

We say that the design matrix $\boldsymbol{X}$ satisfies Restricted Isometry Property (RIP) with order s and constant $0 < \delta_s < 1$ if for all $S \subset [p]$ with $|S| \leq s$ and $\forall u \neq 0, u \in \mathbb{R}^{|S|}$, it holds that

$$1 - \delta_s \leq \frac{\|\boldsymbol{X}_S u\|_2^2}{n\|u\|_2^2} \leq 1 + \delta_s.$$

Denote this condition by $\boldsymbol{X} \sim \mathrm{RIP}(s, \delta_s)$. We present assumptions below to establish the theoretical properties of the HTP algorithm.

Assumption 1: The random errors $\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_n$ are i.i.d. with mean zero and sub-Gaussian tails, that is, there exists a positive number σ such that

$$\mathrm{pr}(|\boldsymbol{\xi}_i| > t) \leqslant 2\exp\left(-\frac{t^2}{2\sigma^2}\right), \quad \text{for all } t \geqslant 0.$$

Assumption 1 helps control the behavior of the statistical error $\boldsymbol{\Xi}$, with further details presented in Lemma 1.

Assumption 2: Design matrix $\boldsymbol{X}$ satisfies $\mathrm{RIP}(3s, \delta_{3s})$.

Assumption 2 serves as a widely used identifiable assumption in the literature on high-dimensional statistics [5], [6], [32]. It requires that all sub-matrices of $\boldsymbol{X}^\top\boldsymbol{X}/n$ with a size smaller than $3s$ exhibit spectral bounds within $[1-\delta_{3s}, 1+\delta_{3s}]$.

Theorem 1 (ℓ_2 error bound): Assume that Assumption 1 and 2 hold. Let $s \geq s^*$ and $\gamma_{3s} := 2\delta_{3s}\sqrt{\frac{1+\delta_{3s}}{1-\delta_{3s}}} < 1$. Then, with probability at least $1 - C_1\exp\{-C_2 s\log(p/s)\}$, we have

$$\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2 \leq \gamma_{3s}^t\|\boldsymbol{\beta}^*\|_2 + \frac{12\sigma}{1-\gamma_{3s}}\left\{\frac{s\log(p/s)}{n}\right\}^{1/2}. \tag{3}$$

Theorem 1 demonstrates that under Assumption 2, the convergence rate of HTP is governed by the contraction factor γ_{3s}. This indicates that the estimation error diminishes geometrically toward the ground truth. Here, γ_{3s} restricts an upper bound for the parameter δ_{3s}, limiting the correlation among the columns of $\boldsymbol{X}$. Furthermore, under the condition of δ_{3s}, $\boldsymbol{\Phi}$ acts as the contraction factor, ensuring a reduction in the optimization error throughout the iterations.

Remark 1: A sufficient condition for $\gamma_{3s} < 1$ is $\delta_{3s} < 0.3478$, which is weaker than the condition $\delta_{3s} < 0.1599$ in [5] and 0.1877 in [6].

Corollary 1: Assume that all the conditions in Theorem 1 hold. Then, we have

$$\mathrm{pr}\left\{\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2 \leq \frac{12\sigma}{1-\gamma_{3s}}\left\{\frac{s\log(p/s)}{n}\right\}^{1/2}\right\}$$
$$\geq 1 - C_1\exp\{-C_2 s\log(p/s)\}$$

for $t \geq \log_{1/\gamma_{3s}}\left\{\frac{(1-\gamma_{3s})\|\boldsymbol{\beta}^*\|_2}{\sigma}\left\{\frac{n}{s\log(p/s)}\right\}^{1/2}\right\}$. Theorem 1 suggests that the estimation error of HTP is influenced by the non-vanishing terms. Following a few iterations, Corollary 1 provides the non-asymptotic upper bounds for the estimation error of HTP. Specifically, when s is of the same order as s^*, the HTP algorithm achieves the minimax optimal rate [33].

C. Adaptation to the Unknown Sparsity and Variance

This section addresses the construction of an adaptive estimator that achieves minimax optimality without prior knowledge of the sparsity s^* and noise level σ^2. In the following development of algorithms, we treat s as a tuning parameter and aim to determine the optimal model size. Given an upper bound $s_{\max}$ of s, this entails running Algorithm 1 along the sequence $[s_{\max}]$, followed by employing a model selection criterion for identifying the optimal model size.

We use the information criterion, which is considered by [26]:

$$\mathrm{IC}(s) = \log\mathcal{L}_n(\hat{\boldsymbol{\beta}}^{(s)}) + K\frac{s}{n}\log\frac{p}{s}, \tag{4}$$

where $\hat{\boldsymbol{\beta}}^{(s)}$ is the estimator of Algorithm 1 given model size s, and K is an absolute constant. The estimator minimizing the criterion (4) is the optimal model size. To get an adaptive minimax optimal estimator, the following assumption is required:

Assumption 3: The sample size n is large enough such that for some positive constant C_0,

$$n \geq C_0 s_{\max}\log\left(p/s_{\max}\right).$$

The literatures [26] and [34] demonstrated the impossibility of simultaneous adaptation to s^* and σ in the ultra-high-dimensional setting, i.e., $n \leq C_0 s^*\log(p/s^*)$. As the upper bound of the tuning parameter s representing s^*, it is natural to require Assumption 3 holds for $s_{\max}$. Hence, Assumption 3 is an almost necessary condition for the adaptation to the unknown sparsity and noise level. It formalizes the fundamental dimensionality-sample size trade-off: while larger $s_{\max}$ allows more flexible adaptation to unknown sparsity patterns, it simultaneously imposes stricter sample size constraints to maintain statistical accuracy. Additionally, Assumption 3 offers guidance on setting an upper bound for the maximum model size $s_{\max}$, which can prevent overfitting meanwhile maintaining computational efficiency. In the numerical experiments, we set $s_{\max} = \lceil n/\log p\rceil$ as an eligible upper bound.

Theorem 2 (Minimax adaptation): Assume that Assumption 1 and 3 hold. Assume that $\boldsymbol{X}$ satisfies $\mathrm{RIP}(3s^*, \delta_{3s^*})$ with $\delta_{3s^*} < 0.3478$. We run Algorithm 1 along sequence $[s_{\max}]$. Denote $\hat{s}$ as the model

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size identified by information criterion (4), and $\hat{\boldsymbol{\beta}}^{(\hat{s})}$ as the corresponding estimator given model size $s = \hat{s}$. Then, we have

$$\mathrm{pr}\,(\hat{s} \le 2s^*) \ge 1 - C_2 \exp\{-C_3 s^* \log(p/s^*)\}, \tag{5}$$

and

$$\mathrm{pr}\left\{\|\hat{\boldsymbol{\beta}}^{(\hat{s})} - \boldsymbol{\beta}^*\|_2 \le C_1\sigma\left\{\frac{s^*\log(p/s^*)}{n}\right\}^{1/2}\right\} \ge 1 - C_2\exp\{-C_3 s^*\log(p/s^*)\}. \tag{6}$$

Theorem 2 stands as a pivotal outcome of our theoretical discoveries. When combined with the information criterion (4), the HTP algorithm achieves minimax optimality with unknown sparsity and variance. Moreover, Theorem 2 indicates that the selected model size $\hat{s}$ can be controlled within $O(s^*)$.

Remark 2: In Theorem 2, we assume that the RIP-type condition holds for parameter $3s^*$ rather than $3s_{\max}$, which imposes a weaker assumption than the Lepski-type method [12]. In particular, we first prove (5) by contradiction and control the selected model size $\hat{s}$ within $2s^*$. Then, based on the result of (5), we only need the RIP condition to hold for parameter $3s^*$.

Remark 3: Criterion (4) can be considered as a variant of the Birgé-Massart criterion [25]. It is important to note that [26] demonstrated that, using criterion (4), an estimator derived from exhaustive search with $s \in [(n-1)/4]$ attains minimax optimality. However, both [25] and [26] provide a theoretical framework to choose s, and do not apply it to a polynomial-time algorithm. HTP efficiently procures high-quality solutions for high-dimensional sparse regression, which makes criterion (4) computationally tractable in practice.

Remark 4: As shown in Theorem 1, achieving optimal statistical performance necessitates that $\hat{s}$ is of the same order as s^*. Following the approach of [12], we can use a geometric grid $\mathcal{G} := \left\{1, 2, 4, \cdots, 2^{\lceil \log_2(s_{\max}) \rceil}\right\}$, instead of running the procedure from 1 to $s_{\max}$ one by one. This setting ensures that the candidate set includes a value of the same order as s^*, and the overall computation cost of the adaptive procedure scales logarithmically with respect to $s_{\max}$, making our procedure more computationally efficient and practically scalable.

III. Oracle Rate Under the Beta-Min Condition

In this section, we establish the theoretical guarantees of HTP under the beta-min condition. Specifically, in Section III-A, we show that HTP can achieve the oracle estimation rate under $\min_{i\in S^*}|\boldsymbol{\beta}_i^*| \ge C\sigma\{\log(p/s^*)/n\}^{1/2}$. Moreover, under a slightly stronger condition $\min_{i\in S^*}|\boldsymbol{\beta}_i^*| \ge C\sigma(\log p/n)^{1/2}$, we demonstrate that the HTP estimator converges to the oracle least-squares estimator with high probability. Section III-B investigates the theory of the minimum signal strength of the HTP estimator with different model sizes. Based on this result, we provide an adaptive tuning procedure to achieve the exact support recovery with unknown sparsity.

A. Oracle Estimation Rate

Consider parameter space $\Omega^p_{s^*,\lambda}$ as

$$\Omega^p_{s^*,\lambda} = \{\boldsymbol{\beta}^* \in \mathbb{R}^p : \|\boldsymbol{\beta}^*\|_0 \le s^* \text{ and } \min_{i\in S^*}|\boldsymbol{\beta}_i^*| \ge \lambda\}.$$

The beta-min condition $\min_{i\in S^*}|\boldsymbol{\beta}_i^*| \ge \lambda$ characterizes the scale of the true signal. Additionally, we demonstrate that if $\min_{i\in S^*}|\boldsymbol{\beta}_i^*| > 2C_1\sigma\{\log(p/s^*)/n\}^{1/2}$, where constant C_1 is determined in (6), the selected model size $\hat{s}$ determined by criterion (4) satisfies

$$\mathrm{pr}\,(s^*/2 \le \hat{s} \le 2s^*) \ge 1 - C_2\exp\{-C_3 s^*\log(p/s^*)\}. \tag{7}$$

The upper bound of $\hat{s}$ has been proved in Theorem 2. We now prove the lower bound of $\hat{s}$ by contradiction. Assume $\hat{s} \le s^*/2$ and denote $\tilde{S} = S^* \setminus \mathrm{supp}(\hat{\boldsymbol{\beta}})$, therefore $|\tilde{S}| \ge s^*/2$ by assumption. Then, we conclude that

$$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2 > \|\hat{\boldsymbol{\beta}}_{\tilde{S}} - \boldsymbol{\beta}^*_{\tilde{S}}\|_2 > C_1\sigma\left\{\frac{s^*\log(p/s^*)}{n}\right\}^{1/2},$$

which leads to a contradiction with (6). Therefore, (7) holds.

Result (7) asserts that, with high probability, HTP with criterion (4) selects a model with the correct order of dimension under the beta-min condition. In addition, (7) implies that $s^* \in [\hat{s}/2, 2\hat{s}]$ with high probability. Consequently, in what follows, we identify the optimal model size within this interval. We only concern the theoretical guarantees of HTP with interval $s \in [s^*, 4s^*]$ rather than $[s^*, \infty)$. Denote

$$\widetilde{\gamma}_{3s,\epsilon} := \sqrt{\frac{1+\delta_{3s}}{1-\delta_{3s}}}\left(1 + 2\sqrt{2} + \frac{3\sqrt{2}}{\epsilon}\right)\delta_{3s} + 8\sqrt{\frac{\delta_{3s}}{1-\delta_{3s}}}.$$

Theorem 3 (Oracle estimation rate): Assume that Assumptions 1, 2, and 3 hold. Assume that $\widetilde{\gamma}_{3s,\epsilon} < 1$ and $\lambda \ge (1+3\epsilon)\sigma\{2\log(p/s^*)/n\}^{1/2}$ for some $0 < \epsilon < 1$. Assume that the initial estimator $\boldsymbol{\beta}^0$ is a minimax optimal estimator, that is, $\|\boldsymbol{\beta}^0 - \boldsymbol{\beta}^*\|_2 \le C_1\sigma\{s^*\log(p/s^*)/n\}^{1/2}$ and $\|\boldsymbol{\beta}^0\|_0 \le s$. Then, when $s \in [s^*, s^* + s^*/\log(p/s^*)]$, for $t \ge \log\left\{C_{1,\epsilon}^2\log(p/s^*)\right\}$, we have

$$\sup_{\boldsymbol{\beta}^*\in\Omega^p_{s^*,\lambda}} \mathrm{pr}\left\{\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2 \ge C_{2,\epsilon}\sigma\left(\frac{s^*}{n}\right)^{1/2}\right\} \le \exp(-C_{3,\epsilon}s^*) + \left(\frac{s^*}{p}\right)^{C_{4,\epsilon}}. \tag{8}$$

Moreover, when $s = s^* + \tau$ where $\tau \in (s^*/\log(p/s^*), 3s^*]$, for $t \ge \log\left(C_1^2 s^*/\tau\right)$, we have

$$\sup_{\boldsymbol{\beta}^*\in\Omega^p_{s^*,\lambda}} \mathrm{pr}\left\{\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2 \ge C_{2,\epsilon}\sigma\left(\frac{\tau\log(p/s^*)}{n}\right)^{1/2}\right\} \le \exp(-C_{3,\epsilon}s^*) + \left(\frac{s^*}{p}\right)^{C_{4,\epsilon}}, \tag{9}$$

where $C_{1,\epsilon}, C_{2,\epsilon}, C_{3,\epsilon}$ and $C_{4,\epsilon}$ are some positive constants related to ϵ.

Theorem 3 demonstrates that when $s \in [s^*, s^* + s^*/\log(p/s^*)]$, HTP estimator attains the oracle estimation rate $O(\sigma(s^*/n)^{1/2})$ after a few iterations with high probability. For $s \in (s^* + s^*/\log(p/s^*), 4s^*]$, the estimation error of HTP gradually diverges from the oracle rate and increases to the minimax optimal rate. Figure 1 illustrates the theoretical results of Theorem 3.

Remark 5: The parameter $\epsilon > 0$ governs the fundamental trade-off between the beta-min condition and the restricted isometry constant δ_{3s}:

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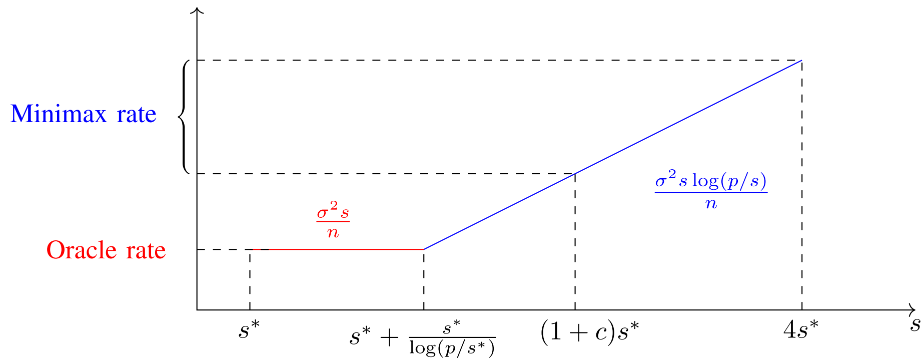

Fig. 1. The estimation error bound of HTP estimator with metric $\|\cdot\|_2^2$ for model size $s \in [s^*, 4s^*]$. Here, c is some positive constant.

- As $\epsilon \to \infty$, the beta-min condition relaxes while the RIP requirement weakens to $\delta_{3s} < 0.21$.
- As $\epsilon \to 0$, the beta-min constant approaches the information-theoretically optimal value $\sqrt{2}$ [15], [20], requiring $\delta_{3s} \to 0$ (i.e., near-identity design).

This characterization preserves theoretical sharpness while accommodating practical design scenarios.

Remark 6: [20] established the oracle estimation rate as

$$\inf_{\hat{\boldsymbol{\beta}}} \sup_{\boldsymbol{\beta}^* \in \Omega^p_{s^*,\lambda}} \mathbf{E}_{\hat{\boldsymbol{\beta}}} \|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2^2 \asymp \frac{\sigma^2 s^*}{n},$$
$$\forall \lambda \geq (1+\epsilon)\sigma\{2\log(p/s^*)/n\}^{1/2}. \tag{10}$$

where $\mathbf{E}_{\hat{\boldsymbol{\beta}}}$ represents the expectation with respect to $\hat{\boldsymbol{\beta}}$. As a result, Theorem 3 and (10) together show that HTP achieves the minimax optimality under parameter space $\Omega^p_{s^*,\lambda}$.

Corollary 2: Assume that all the conditions in Theorem 3 hold. Then, when $s \in [s^*, s^* + s^*/\log(p/s^*)]$, for $t \geq \log\left\{C_{1,\epsilon}^2 \log(p/s^*)\right\}$, the type-I and type-II errors of variable selection satisfy

$$\sup_{\boldsymbol{\beta}^* \in \Omega^p_{s^*,\lambda}} \mathrm{pr}\left\{\frac{|S^t \Delta S^*|}{s^*} = \Omega\left(\frac{1}{\log(p/s^*)}\right)\right\}$$
$$\leq \exp(-C_{2,\epsilon} s^*) + \left(\frac{s^*}{p}\right)^{C_{3,\epsilon}}.$$

Corollary 2 shows that, when the estimator achieves the oracle rate, both the type-I and type-II errors of variable selection can be controlled within $o(s^*)$ with high probability.

Remark 7: Figure 2 shows the iterative thresholding dynamics of the estimator of HTP. Given an initial solution and letting $s = s^*$, the IHT estimator first hits the minimax optimal region, i.e., the blue region in Figure 2. Then, as t increases, the estimator achieves the oracle rate, i.e., the red region, and the error of variable selection can be controlled at a low level, as Corollary 2 shows. This behavior benefits from dynamic signal detection, as shown by [15]. Under the beta-min condition, the support recovery result enables HTP to exceed the minimax rate and attain the oracle rate.

Theorem 3 and Corollary 2 illustrate that the HTP estimator performs as well as the oracle estimator on estimation asymptotically, although it may not necessarily recover the true support set S^* exactly. By strengthening the beta-min condition in Theorem 3, Theorem 4 demonstrates that the HTP estimator converges to the oracle estimator with high probability. Consequently, Corollary 2 is strengthened as Corollary 3, affirming that the HTP estimator can exactly recover the true support set S^* with high probability. Denote the oracle least-squares estimator $\tilde{\boldsymbol{\beta}}^*$ as

$$\tilde{\boldsymbol{\beta}}^*_{S^*} = (\boldsymbol{X}_{S^*}^\top \boldsymbol{X}_{S^*})^{-1} \boldsymbol{X}_{S^*}^\top \boldsymbol{y}, \text{ and } \tilde{\boldsymbol{\beta}}^*_{(S^*)^c} = 0.$$

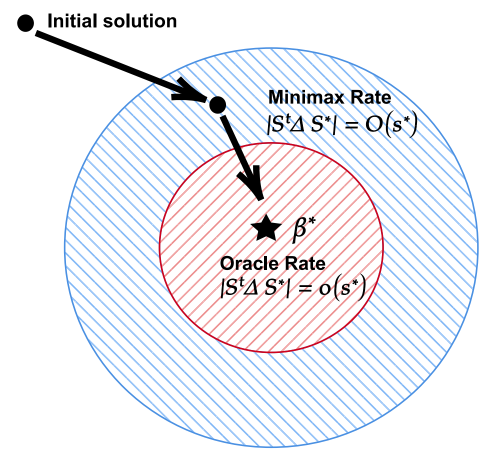

Fig. 2. The solution path of HTP as iteration t increases. The blue region indicates the iterations achieving the minimax rate, while the red region represents the iterations reaching the oracle rate.

In the following theorem, we establish the error bounds between the HTP estimator and the oracle estimator.

Theorem 4: Assume that all the conditions in Theorem 3 hold and $\lambda \geq (2+2\epsilon)\sigma(2\log p/n)^{1/2}$. Then, given $s = s^* + \tau$ where $\tau \in [0, 3s^*]$, we have

$$\sup_{\boldsymbol{\beta}^* \in \Omega^p_{s^*,\lambda}} \mathrm{pr}\Bigg\{\|\boldsymbol{\beta}^t - \tilde{\boldsymbol{\beta}}^*\|_2 \geq \widetilde{\gamma}^t_{3s,\epsilon} C_{1,\epsilon}\sigma\left(\frac{s^*\log p}{n}\right)^{1/2} + C_{2,\epsilon}\sigma\left(\frac{2\tau\log p}{n}\right)^{1/2}\Bigg\} \leq \left(\frac{s^*}{p}\right)^{C_{3,\epsilon}}.$$

For $s > s^*$, Theorem 4 demonstrates that the estimation error rate is $O(\sigma((s - s^*)\log p/n)^{1/2})$. Notably, when $s = s^*$, the error diminishes to zero with iterations, implying that the HTP estimator converges to the oracle estimator with a probability approaching 1 as $s^*/p \to 0$.

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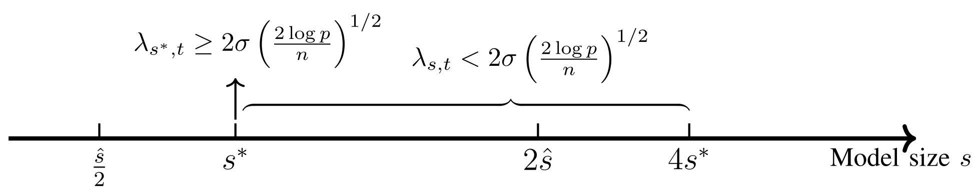

Fig. 3. The minimum signal strength of estimator $\boldsymbol{\beta}^t$ for different model sizes under the conditions of Theorem 5.

Corollary 3 (Exact support recovery): Assume that all the conditions in Theorem 4 hold. Then, when $s = s^*$ and $t \geq 2\log\left((C_2/\epsilon)^2 s^*\right)$, we have

$$\lim_{s^*/p\to 0} \sup_{\boldsymbol{\beta}^* \in \Omega^p_{s^*,\lambda}} \mathrm{pr}\left(S^t = S^*\right) = 1.$$

Corollary 3 shows that given $s = s^*$, HTP algorithm can recover the true support set S^* exactly in finite iterations with high probability.

Remark 8: Existing exact support recovery results for ℓ_1 estimators [35], [36] relied heavily on the irrepresentable condition, recognized as a stronger condition compared to the RIP [37]. Typically, achieving exact support recovery demands a tight ℓ_∞ error bound, and in this context, incoherence conditions become inevitable. Some studies circumvent these requirements by leveraging the RIP condition, establishing surrogate ℓ_2 error bounds instead. However, this approach often imposes a more stringent beta-min condition [4], [5], [6], [38], [39]. In our work, we establish the ℓ_∞ error bound by asymptotically connecting the oracle ordinary least-squares estimator. Specifically, Theorem 4 shows that the HTP estimator asymptotically converges to the oracle estimator when $s = s^*$. Consequently, we can establish a tight ℓ_∞ error bound by solely using the RIP condition. Therefore, HTP attains the exact support recovery with the optimal beta-min condition and RIP condition.

B. Adaptation to the Exact Support Recovery

In this section, we consider an adaptive tuning procedure to identify the optimal model size. Denote $\lambda_{s,t}$ as the smallest nonzero value of $|\boldsymbol{\beta}^t|$ given model size s, that is, the minimum magnitude of the nonzero estimate of signal at step t in Algorithm 1. In the following theorem, we show that after a few iterations, the minimum signal strength of the estimator $\boldsymbol{\beta}^t$ exhibits a separable phenomenon for specific model sizes.

Theorem 5: Assume that the conditions in Theorem 3 hold and $\lambda \geq \left(C_1 \vee 4\right)\sigma(2\log p/n)^{1/2}$, where constant C_1 is determined in (6). Given $s = s^*$, for $t \geq 2\log(C_2^2 s^*)$, we have

$$\sup_{\boldsymbol{\beta}^* \in \Omega^p_{s^*,\lambda}} \mathrm{pr}\left\{\lambda_{s^*,t} < \left((C_1 - 2) \vee 2\right)\sigma\left(\frac{2\log p}{n}\right)^{1/2}\right\} \leq \left(\frac{s^*}{p}\right)^{C_3}. \tag{11}$$

On the other hand, given $s \in (s^*, 4s^*]$, for $t \geq 2\log(C_2^2 s^*)$, we have

$$\sup_{\boldsymbol{\beta}^* \in \Omega^p_{s^*,\lambda}} \mathrm{pr}\left\{\lambda_{s,t} \geq 2\sigma\left(\frac{2\log p}{n}\right)^{1/2}\right\} \leq \left(\frac{s^*}{p}\right)^{C_3}. \tag{12}$$

Theorem 5 demonstrates that when the size s matches the true sparsity s^*, the minimum signal strength becomes adequately substantial after $t \geq 2\log(C_2^2 s^*)$. Additionally, for cases where $s > s^*$, the minimum signal strength can be effectively controlled below the threshold of $2\sigma(2\log p/n)^{1/2}$. This relationship is visually depicted in Figure 3.

To determine the optimal size s^*, we suggest an adaptive tuning procedure. A straightforward approach is to apply hard thresholding with a threshold of $2\sigma(2\log p/n)^{1/2}$ to the minimum signal strength of estimator $\boldsymbol{\beta}^t$ with model size belonging to $[\hat{s}/2, 2\hat{s}]$. The largest model size for which the minimum signal strength of the estimator $\boldsymbol{\beta}^t$ exceeds the threshold $2\sigma(2\log p/n)^{1/2}$ indicates the optimal model size s^*. We call this the adaptive tuning procedure. It can successfully identify the optimal model size s^* with high probability. Combining with Theorem 4 and Corollary 3, we prove the selection consistency of the HTP estimator. The HTP estimator adaptively converges to the oracle estimator with high probability.

Remark 9: The optimal threshold of the adaptive tuning procedure relies on the true noise level σ. However, the noise level σ is usually unknown in practice. Fortunately, we can obtain a reliable estimate using the minimax optimal estimator $\hat{\boldsymbol{\beta}}^{(\hat{s})}$ shown in Theorem 2, even in high-dimensional settings. Specifically, we can replace σ in the threshold with the plug-in estimator $\hat{\sigma} = \|\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{(\hat{s})}\|_2/n^{1/2}$ to facilitate optimal model size selection.

Remark 10: To enhance its empirical performance, we propose an approach that involves comparing the ratios of the minimum signal strengths of adjacent estimators. By employing a pre-determined threshold κ, we identify the optimal size by determining the maximum index where the ratio exceeds κ. In our numerical experiments, we set κ to 2. Appendix L demonstrates the robustness of this parameter choice through extensive numerical experiments.

In particular, the beta-min condition is necessary to achieve the separation of the minimum signal strength of HTP estimators as shown in Theorem 5. However, in practical applications, verifying the beta-min condition might be challenging. Hence, evaluating the effectiveness of the adaptive tuning procedure becomes crucial. Let $\tilde{s}$ be the selected sparsity by the adaptive tuning procedure, and $\hat{s}$ be the selected sparsity in Section II-C. To assess its efficiency, we examine whether the following condition holds:

$$\left\|\hat{\boldsymbol{\beta}}^{(\tilde{s})} - \hat{\boldsymbol{\beta}}^{(\hat{s})}\right\|_2^2 \leq C\sigma^2 \frac{\hat{s}\log(p/\hat{s})}{n}.$$

If the above condition is met, we conclude that the adaptive tuning procedure is efficient. This means that our adaptive tuning procedure is signal-adaptive: when the beta-min

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condition in Theorem 5 is not satisfied, the estimation error of $\hat{\boldsymbol{\beta}}^{(\tilde{s})}$ has the minimax rate $\sigma\{s^*\log(p/s^*)/n\}^{1/2}$. On the other hand, when the beta-min condition is fulfilled, the estimation error adaptively meets the oracle rate $\sigma(s^*/n)^{1/2}$. We summarize the above observation as the following corollary:

Corollary 4: Given two estimators $\hat{\boldsymbol{\beta}}^{(\tilde{s})}$ and $\hat{\boldsymbol{\beta}}^{(\hat{s})}$ where $\hat{s} = \arg\min_{s\in[s_{\max}]} \mathrm{IC}(s)$, and $\tilde{s}$ is the selected model size by the adaptive tuning procedure. Define an estimator

$$\hat{\boldsymbol{\beta}} = \begin{cases} \hat{\boldsymbol{\beta}}^{(\tilde{s})} & \text{, if } \left\|\hat{\boldsymbol{\beta}}^{(\tilde{s})} - \hat{\boldsymbol{\beta}}^{(\hat{s})}\right\|_2^2 \le C\sigma^2 \frac{\hat{s}\log(p/\hat{s})}{n}, \\ \hat{\boldsymbol{\beta}}^{(\hat{s})} & \text{, otherwise,} \end{cases}$$

where $C = 2(5+C_1)^2$ and C_1 is determined in (6). Assume that all conditions in Theorem 5 hold. Then, as $s^*, p/s^* \to \infty$, with probability tending to 1, the estimator $\hat{\boldsymbol{\beta}}$ achieves adaptivity to the minimum signal strength, that is,

$$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2 \le \begin{cases} 5\sigma\left(\frac{s^*}{n}\right)^{1/2} & \text{, if } \boldsymbol{\beta}^* \in \Omega^p_{s^*,\lambda}, \\ \widetilde{C}\sigma\left\{\frac{s^*\log(p/s^*)}{n}\right\}^{1/2} & \text{, otherwise,} \end{cases}$$

where $\lambda = (C_1 \vee 4)\sigma\left(\frac{2\log p}{n}\right)^{1/2}$ and $\widetilde{C} = (2C)^{1/2} + C_1$.

This corollary shows that our method attains a similar phase transition in [20], demonstrating the minimax rate-optimal signal adaptivity of FAHTP. In contrast, even under stronger conditions, the LASSO estimator cannot achieve the sharper rate $\sigma(s^*/n)^{1/2}$ [21], and the Adaptive LASSO [38] and the Thresholded LASSO [39] also fail to do so.

Consequently, by Corollary 4 and Theorem 4 in [40], we obtain the asymptotic property of the adaptive tuning HTP estimator, as shown in the following.

Corollary 5 (Asymptotic properties): Assume that the conditions in Corollary 4 hold. When $\boldsymbol{\beta}^* \in \Omega^p_{s^*,\lambda}$, the estimator $\hat{\boldsymbol{\beta}}$ defined in Corollary 4 satisfies

$$\lim_{n,s^*\to\infty,\ s^*/p\to 0} \mathrm{pr}\left(\hat{\boldsymbol{\beta}} = \tilde{\boldsymbol{\beta}}^*\right) = 1.$$

In addition, assume that $s^* = o(n^{1/3})$, $E|\boldsymbol{\xi}_i|^3 \le C_1\sigma^3$ and $|\boldsymbol{X}_{ij}| \le C_2$ for all $i \in [n]$, $j \in S^*$. Then, for all $\boldsymbol{a} \in \mathbb{R}^{s^*}$, the asymptotic distribution is

$$\sqrt{n}\cdot \boldsymbol{a}^\top(\hat{\boldsymbol{\beta}}_{S^*} - \boldsymbol{\beta}^*_{S^*}) \to \mathcal{N}\left(0, \sigma^2\boldsymbol{a}^\top\hat{\boldsymbol{\Sigma}}^{-1}_{oracle}\boldsymbol{a}\right),$$

where $\hat{\boldsymbol{\Sigma}}_{oracle} = \boldsymbol{X}_{S^*}^\top\boldsymbol{X}_{S^*}/n$.

Therefore, the estimator $\hat{\boldsymbol{\beta}}$ in Corollary 4 possesses the strong property when the beta-min condition is met. Here we set constant $C = 5$ and provide the full-adaptive version of the HTP algorithm in Algorithm 2.

Algorithm 2 Full-Adaptive HTP (FAHTP) Algorithm

Require: $\boldsymbol{X}$, $\boldsymbol{y}$, $s_{\max}$ and κ.
for $s = 1, \ldots, s_{\max}$, **do**
  $\hat{\boldsymbol{\beta}}^{(s)} = \mathrm{HTP}(\boldsymbol{X}, \boldsymbol{y}, s)$.
  $\lambda_{\min}(s) = \min_{i\in\mathrm{supp}(\hat{\boldsymbol{\beta}}^{(s)})} |\hat{\boldsymbol{\beta}}^{(s)}_i|$.
  Compute the information criterion (4) as IC_s
end for // The first stage
$\hat{s} = \mathrm{argmin}_{s\in[s_{\max}]}\{\mathrm{IC}_s\}$ and compute $\hat{\sigma} = \|\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{(\hat{s})}\|_2/n^{1/2}$.
for $\tilde{s} = 2\hat{s}, \ldots, \hat{s}/2$ **do**
  if $\lambda_{\min}(\tilde{s})/\lambda_{\min}(\tilde{s}+1) \ge \kappa$ and $\|\hat{\boldsymbol{\beta}}^{(\tilde{s})} - \hat{\boldsymbol{\beta}}^{(\hat{s})}\|_2^2 \le 5\hat{\sigma}^2\hat{s}\log(p/\hat{s})/n$, **then**
    Break and let $\hat{s} = \tilde{s}$.
  end if
end for // The second stage
Ensure: $\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^{\hat{s}}$.

IV. Analysis of the General Signals

The previous theoretical guarantees are derived under the assumption of the beta-min condition. However, the beta-min condition may not hold in practice. Therefore, analyzing the error bounds for general signals, i.e., without assuming the beta-min condition, is of particular interest. [41] established a tighter error bound exceeding the minimax rate through implicit regularization with an early stopping criterion. They divided the signals into strong and weak parts, demonstrating that achieving a tighter bound hinges on accurately identifying the number of strong signals and controlling the total magnitude of the weak signals.

In this section, we divide the support set S^* into $S_1^* := \{i \in S^* : |\boldsymbol{\beta}_i^*| \ge C_1\sigma(\log p/n)^{1/2}\}$ and $S_2^* := \{i \in S^* : |\boldsymbol{\beta}_i^*| < C_1\sigma(\log p/n)^{1/2}\}$, which correspond to the sets of strong and weak signals, respectively. We can derive a generalized version of Theorem 3 as follows:

Theorem 6: Assume that the conditions in Theorem 3 hold. Then, for the above general signals, given $s = |S_1^*| + \tau$, for $t \ge \log\left(C_2^2\log p\right)$, we have

$$\mathrm{pr}\left\{\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2^2 \ge C_3\left(\sigma^2\frac{|S_1^*|}{n} + \|\boldsymbol{\beta}^*_{S_2^*}\|_2^2\right)\right\} \le \exp(-C_4|S_1^*|) + \left(\frac{|S_1^*|}{p}\right)^{C_5}. \tag{13}$$

For the general signals, Theorem 6 characterizes a tighter error bound than the minimax rate. It demonstrates that the upper bound can be decomposed into two components: the error associated with estimating the strong signals and the error related to the weak signals. Notably, the first component is independent of the dimension p, while the second can be controlled by the total magnitude of the weak signals. A similar result has been achieved through implicit regularization with an early stopping criterion [41]. Specifically, when $s^* = |S_1^*|$, we recover the result (8) in Theorem 3.

Corollary 6: Assume the conditions in Theorem 6 hold and $(|S_1^*|\log p)/n \to 0$, $|S_1^*|/p \to 0$ and $|S_1^*| \to \infty$. Then, by Algorithm 2, with probability tending to 1, we have

$$\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2^2 \le \begin{cases} C_2\left(\sigma^2\frac{|S_1^*|}{n} + \|\boldsymbol{\beta}^*_{S_2^*}\|_2^2\right) & , \|\boldsymbol{\beta}^*_{S_2^*}\|_2 \le C_1\sigma\left(\frac{\log p}{n}\right)^{1/2} \\ C_2\left(\sigma^2\frac{|S_1^*|\log p}{n} + \|\boldsymbol{\beta}^*_{S_2^*}\|_2^2\right) & , \|\boldsymbol{\beta}^*_{S_2^*}\|_2 > C_1\sigma\left(\frac{\log p}{n}\right)^{1/2}. \end{cases}$$

Corollary 6 shows that our proposed FAHTP can achieve a tighter estimation error bound than the minimax rate $\sigma^2 s^*\log p/n$ with high probability. It is worth noting that the Corollary 6 can cover the case of ℓ_q-sparisty ($0 < q < 1$), which implies that FAHTP is minimax adaptive to the unknown q and radius R_q of the ℓ_q-ball [33].

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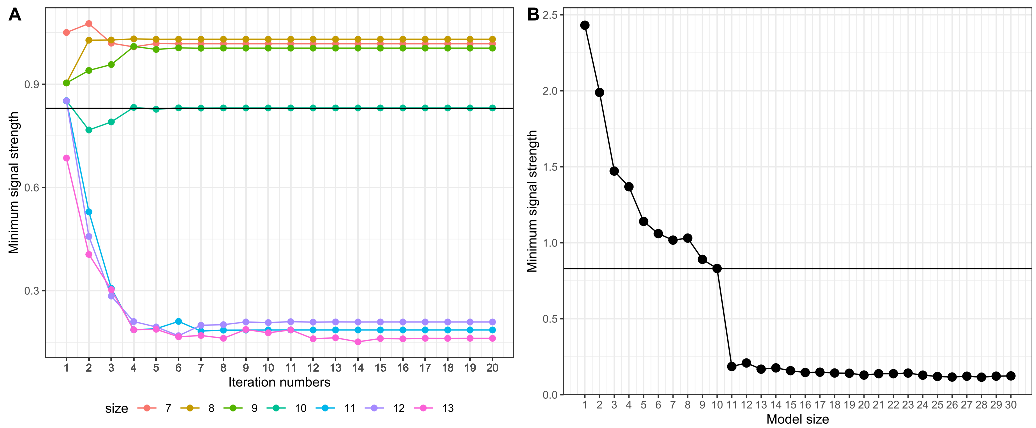

Fig. 4. The phenomenon of the minimum signal strength of HTP estimators with varying model sizes. The horizontal black line represents the true minimal signal strength. (A) The convergence of the minimal strength of estimators within 20 iterations. (B) The minimum signal strength of estimators with different model sizes.

V. Numerical Experiments

This section provides numerical experiments complementing our theoretical findings and offering insights into the empirical performance of our proposed methods. We begin by demonstrating the convergence phenomenon of the minimum signal strength in Section V-A. Following this, Section V-B highlights the efficacy of the adaptive tuning procedure under assumptions on the minimum signal strength. Lastly, we conduct a comparative analysis between our proposed FAHTP algorithm and several state-of-the-art methods by synthetic and real datasets. Here we fix $K = 3$ in the information criterion (4). An extensive exploration of the choosing of K is provided in Appendix K.

Given estimator $(\hat{\boldsymbol{\beta}}, \hat{S})$, we consider five metrics to measure the efficacy. The first one is the estimation error (EE): $\|\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}^*\|_2$. We measure the difference between $|\hat{S}|$ and true sparsity s^* by sparsity error (SE): $|\hat{S}| - |S^*|$. To evaluate the performance of variable selection, we consider the true positive rate (TPR), false positive rate (FPR), and Mathew's correlation coefficient (MCC). Here MCC is defined as

$$\mathrm{MCC} = \frac{\mathrm{TP} \times \mathrm{TN} - \mathrm{FP} \times \mathrm{FN}}{\{(\mathrm{TP}{+}\mathrm{FP})(\mathrm{TP}{+}\mathrm{FN})(\mathrm{TN}{+}\mathrm{FP})(\mathrm{TN}{+}\mathrm{FN})\}^{1/2}},$$

where $\mathrm{TP} = |\hat{S} \cap S^*|$ and $\mathrm{TN} = |\hat{S}^c \cap (S^*)^c|$ stand for true positives/negatives, respectively. $\mathrm{FP} = |\hat{S} \cap (S^*)^c|$ and $\mathrm{FN} = |\hat{S}^c \cap S^*|$ stand for false positives/negatives, respectively. Moreover, in the subsection comparing our method with other advanced methods, we provide a detailed runtime analysis for each algorithm. Specifically, 'Runtime(s)' denotes the execution time in seconds, while 'Runtime(min)' indicates the runtime in minutes.

A. The Convergence of the Minimum Signal Strength

We set $n = 300$, $p = 1000$, $s^* = 10$, and $\sigma = 1$. Each entry of $\boldsymbol{X} \in \mathbb{R}^{n \times p}$ is independently drawn from $\mathcal{N}(0,1)$. The non-zero entries of $\boldsymbol{\beta}^*$ are independently drawn from $\mathrm{Unif}\left\{2\sigma(2\log p/n)^{1/2}, 10\sigma(2\log p/n)^{1/2}\right\}$. Subsequently, we run Algorithm 1 with model sizes ranging from 1 to 30. To investigate the convergence phenomenon of the minimum signal strength for each model size, we maintain a fixed iteration number of 20.

Plot A in Figure 4 reveals that after several iterations, all strengths converge. Particularly, when $s = s^*$, the minimum signal strength fluctuates around the true value and eventually converges to it. Moreover, for $s < s^*$, the minimum signal strength surpasses the true value, whereas for $s > s^*$, it notably falls below the true value. Plot B comprehensively illustrates this phenomenon. Notably, with the assumption regarding the minimal signal strength, there exists a significant gap between model sizes $s = s^*$ and $s > s^*$, aligning with the findings in Theorem 5. This observation indicates the feasibility of applying our adaptive tuning procedure in practice.

B. The Benefit of the Adaptive Tuning Procedure

In this section, we demonstrate the efficacy of the adaptive tuning procedure through simulations. We set $n = 300$, $p = 2000$, $s^* = 30$, $\sigma = 1$, and each entry of $\boldsymbol{X} \in \mathbb{R}^{n\times p}$ is independently drawn from $\mathcal{N}(0,1)$. To control the minimum strength of the true signal, the nonzero entries of $\boldsymbol{\beta}^*$ are independently drawn from $\mathrm{Unif}\left\{k/4 \cdot \sigma(2\log p/n)^{1/2}, 4\sigma(2\log p/n)^{1/2}\right\}$, where k increases from 1 to 16. The parameter k controls the minimal signal strength, with a larger k resulting in a stronger minimum signal strength. Here we consider three methods to estimate the signals:

- **The IC method**: Use information criterion (4) to identify $\hat{s}$.
- **The adaptive tuning method**: Run Algorithm 2.
- **The oracle estimator**: The OLS estimator on the true support set S^*.

The computational results are shown in Figure 5.

From Figure 5, it's evident that for small minimum signal strengths, i.e., $k \leq 4$, both the IC and adaptive tuning methods perform similarly, showing significant discrepancies from the oracle estimators. As the minimum signal strength increases, the adaptive tuning procedure demonstrates its superiority. It exhibits improved performance in both estimation and variable selection compared to the IC method. Specifically, the adaptive tuning procedure effectively limits false positives,

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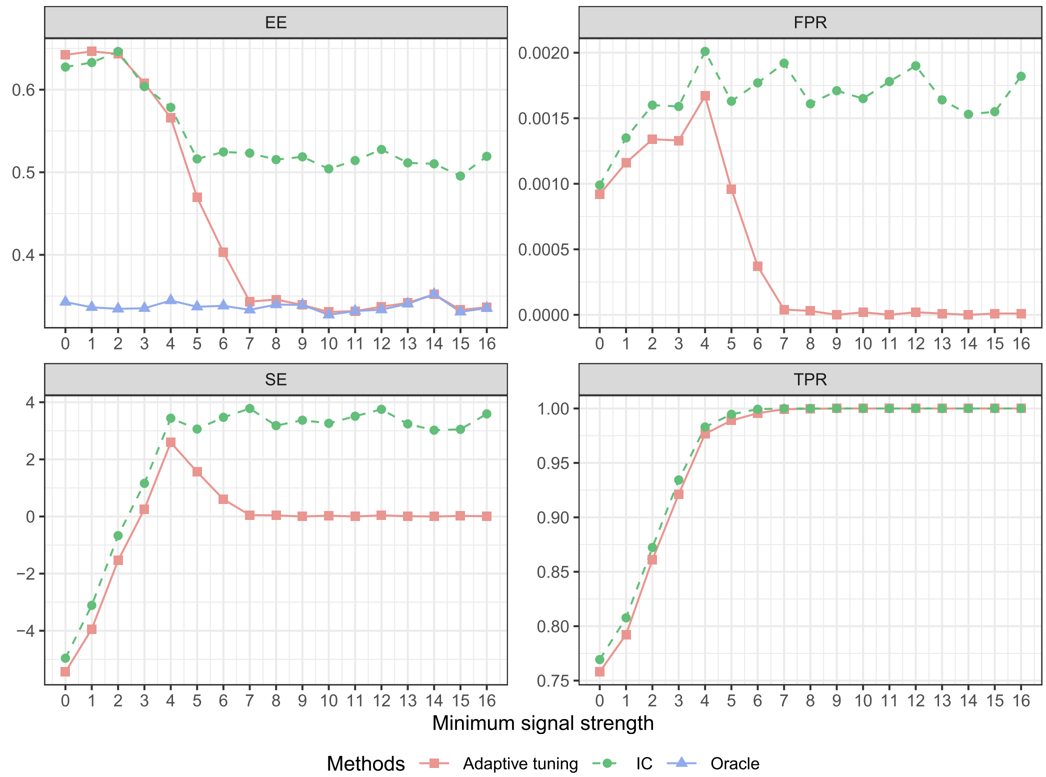

Fig. 5. Performance with increasing minimal signal strength.

resulting in excellent variable selection performance. When the minimum signal strength reaches a sufficiently high level, the adaptive tuning procedure facilitates HTP in achieving the exact recovery of the true support set. Remarkably, the estimation error of the adaptive tuning method matches that of the oracle estimator, aligning with our theoretical findings.

C. Comparison With the State-of-the-Art Methods

We conduct a comparative analysis involving our proposed FAHTP algorithm and several methods: LASSO [17], Adaptive LASSO [42] and Thresholded LASSO [39], fitted using the R package glmnet [43], and SCAD [44] and MCP [45], computed using the R package ncvreg [46]. For parameter tuning across all five methods, we utilize 10-fold cross-validation. Additionally, SCAD and MCP use information criterion (4) to select the optimal tuning parameters.

We set $p = 2{,}000, s^* = 30$ and increase n from 300 to 1,300 with an increment equal to 100. $\boldsymbol{X}$ is generated from the multivariate normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{\Sigma})$, where $\mathbf{\Sigma} = 0.5^{|i-j|}$ for $i, j \in [p]$. $\boldsymbol{\beta}^*$ are uniformly generated from $[1, 5]$ with a random sign. We fix the signal-to-noise ratio (SNR) as 10 to control the noise level σ, where $\text{SNR} = (\boldsymbol{\beta}^*)^\top \mathbf{\Sigma} \boldsymbol{\beta}^* / \sigma^2$. The computational results are shown in Figure 6.

In terms of estimation error, FAHTP exhibits similar performance to SCAD-CV and MCP-CV when the sample size is small, with all methods displaying substantial deviations from the oracle estimator. However, as the sample size increases, FAHTP notably outperforms the others. It is worth noting that in this scenario, FAHTP achieves the same estimation error as the oracle estimator. This observation underscores

TABLE I
THE MEASURES OF IMAGE RECOVERY PERFORMANCE

Method	$\left\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\right\|_2$	PSNR($\hat{\boldsymbol{\beta}}$)	Runtime(min)
FAHTP	**0.075**	**94.172**	**3.352**
LASSO	9.373	52.234	4.812
Adaptive LASSO	6.985	54.787	7.031
Thresholded LASSO	7.303	54.402	16.832
SCAD	0.160	87.601	8.006
MCP	0.121	90.010	9.115

the convergence of FAHTP to the oracle estimator under the assumption of minimum signal strength. Furthermore, metrics such as SE and MCC demonstrate that FAHTP can exactly recover the true support set with a sufficiently large sample size, reinforcing our theoretical findings. As discussed in our introduction, while convex methods like LASSO achieve minimax optimality, their estimator bias prevents scaled optimality. Our numerical experiments confirm this phenomenon, showcasing the superiority of FAHTP over convex methods in both theoretical and practical applications. Notably, FAHTP exhibits the fastest runtime across all compared algorithms, highlighting its computational advantages. Additional simulation results can be found in Appendix M and N.

D. Image Reconstruction Under Noisy Compressed Sensing

This section illustrates how we apply our algorithm to an image recovery task. Following the setup in [47], we transform a 256×256-pixel Shepp-Logan phantom into a $p = 65{,}536$-dimensional coefficient vector $\boldsymbol{\beta}^*$ via a four-level Daubechies

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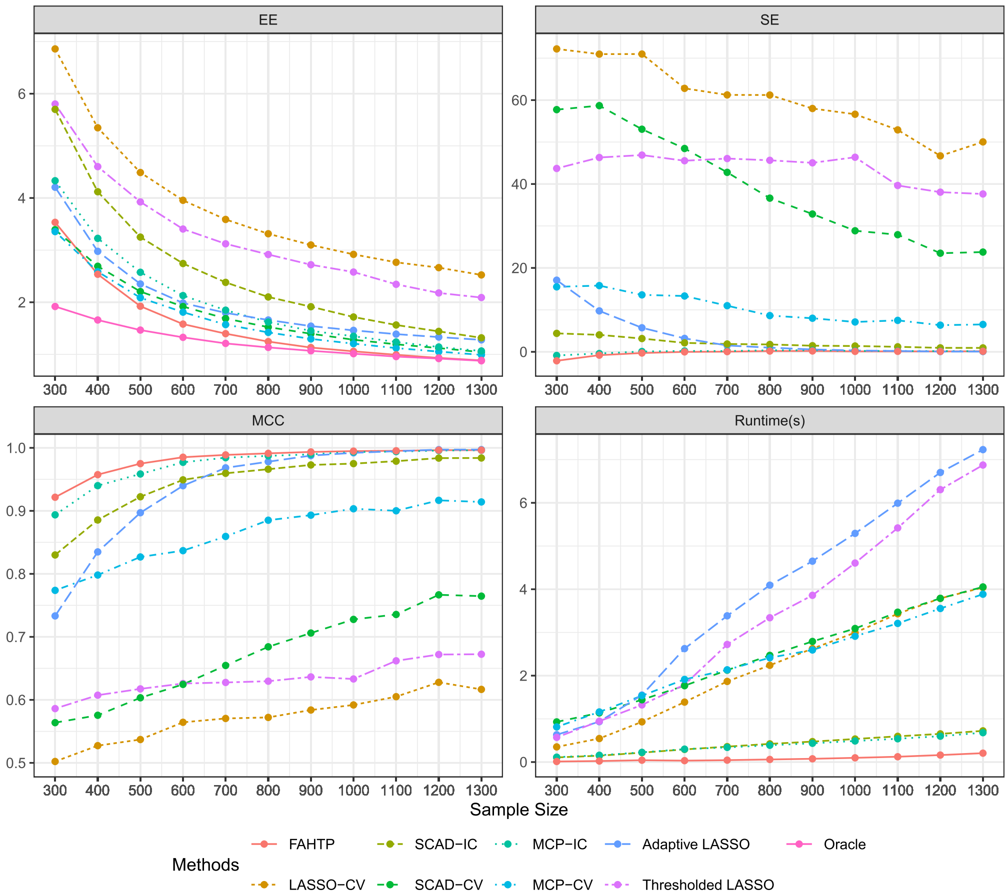

Fig. 6. Performance metrics with an increasing sample size with 100 repetitions. The suffix "-CV" indicates parameter tuning through 10-fold cross-validation. The suffix "-IC" indicates parameter tuning through information criterion (4).

TABLE II

THE COMPUTATIONAL RESULTS OF EACH METHOD BASED ON 200 RANDOM PARTITIONS. THE STANDARD DEVIATION IS SHOWN IN PARENTHESES

Method	Model size	MSE	Runtime(s)
FAHTP	2.33(0.53)	**0.410(0.109)**	**0.148(0.056)**
LASSO-CV	17.51(4.56)	0.425(0.090)	1.773(0.211)
Adaptive LASSO	12.38(3.96)	0.416(0.102)	1.711(0.208)
Thresholded LASSO	8.71(2.64)	0.422(0.113)	1.483(0.195)
SCAD-IC	8.21(1.10)	0.435(0.119)	0.320(0.040)
SCAD-CV	28.89(15.24)	0.419(0.109)	7.311(0.732)
MCP-IC	3.90(7.55)	0.423(0.117)	0.282(0.039
MCP-CV	8.05(6.57)	0.418(0.109)	4.920(0.464)

1 wavelet transformation. The signal $\boldsymbol{\beta}^*$ is $s^* = 3,724$-sparse, leading $s^*/p = 0.057$. We draw the design matrix $\boldsymbol{X} \in \mathbb{R}^{n\times p}$ with observation number $n = \lceil 0.2 \cdot p\rceil = 13,108$, and each entry of $\boldsymbol{X}$ is drawn from a normal distribution independently. Then, we generate the linear model $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta}^* + \boldsymbol{\xi} \in \mathbb{R}^n$, with each $\boldsymbol{\xi}_i \sim \mathcal{N}(0, \sigma^2)$ independently and $\sigma = 10^{-3}$. Our procedure follows the grid search method as mentioned in Remark 4.

Figure 7 shows the ground-truth image and reconstructions from each method, while Table I reports the ℓ_2 errors and the Peak Signal-to-Noise Ratio (PSNR), which is of the form

$$\mathrm{PSNR}(\hat{\boldsymbol{\beta}}) := 10 \cdot \log_{10}\left(\frac{\max_{j\in[p]}\{(\hat{\boldsymbol{\beta}}_j)^2 \vee (\boldsymbol{\beta}_j^*)^2\}}{\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2^2/p}\right).$$

Higher PSNR indicates a closer match to the original image and therefore a better recovery quality. Figure 7 and Table I demonstrate that FAHTP delivers the highest-quality image reconstruction among the six methods.

E. Supermarket Dataset Analysis

We illustrate the performance of each method by a supermarket dataset [48], which consists of 464 records and 6,398 predictors. Each record represents a daily observation from a major supermarket in northern China, with each predictor capturing the sales volume of a specific product. The response variable is the daily customer count. Our goal is to identify

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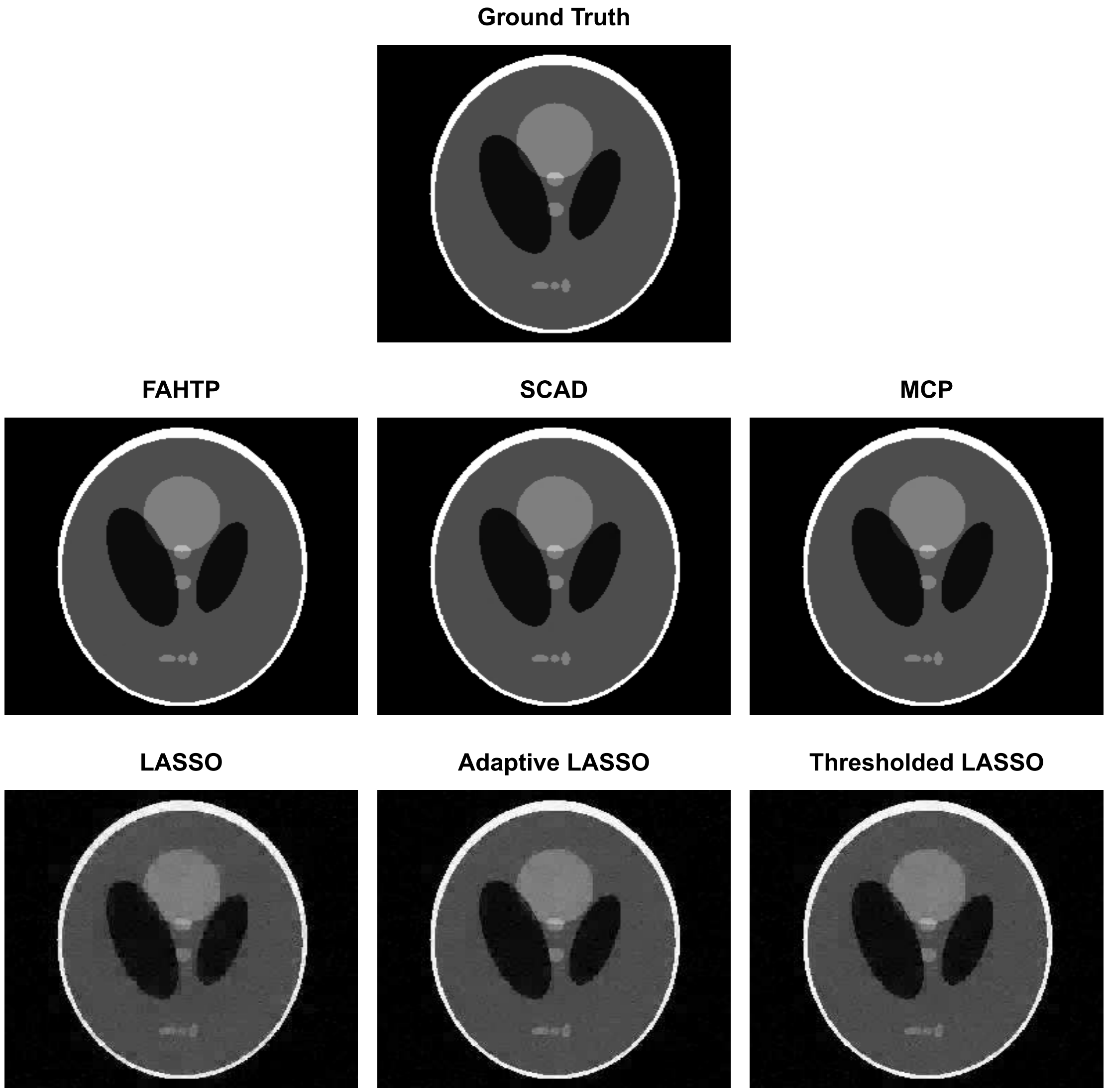

Fig. 7. Reconstruction results of the Shepp-Logan phantom.

which product sales volumes are most strongly correlated with customer numbers.

We conducted 200 random partitions of the dataset. For each partition, 80% of the records are randomly selected as training data, and the remaining 20% as the test set. Each method is fitted on the training set, with its performance evaluated on the test set. Table II summarizes the average model size and the mean square error (MSE) on the test set. We observe that our method tends to select a sparser model with competitive and even outperformed predictive performance compared with the state-of-the-art methods. Additionally, our methods achieves the fastest runtime among all methods, demonstrating its superior computational efficiency.

VI. Conclusion and Discussion

HTP is a classical and widely used method for high-dimensional sparse linear regression. However, some theories, such as how to choose the sparsity s^*, need to be further studied. This study contributes to the minimax optimality and adaptivity of the HTP algorithm. In this paper, we propose a novel tuning-free method, named FAHTP, to simultaneously adapt to the unknown sparsity s^*, noise level σ, and signal strength. Our procedure ensures the minimax rate $\sigma^2 s^* \log(p/s^*)/n$ first and can achieve the oracle estimation rate $\sigma^2 s^*/n$ under optimal beta-min condition. In other words, we have fully extended the good performance of the hard thresholding estimator in Gaussian sequence models to linear regression, and the assumption on the design matrix solely depends on the Restricted Isometric Property. More importantly, even without the beta-min conditions, our FAHTP achieves a tighter error bound than the minimax rate with high probability.

Notably, this work, especially the conclusion for oracle estimation rate and support recovery, reveals an intriguing aspect of HTP when compared to convex estimators like LASSO, which has been proven that under any signal conditions, it is impossible to break through the minimax optimal rate. Therefore, we theoretically demonstrate some numerical phenomena observed in previous studies where the HTP algorithm

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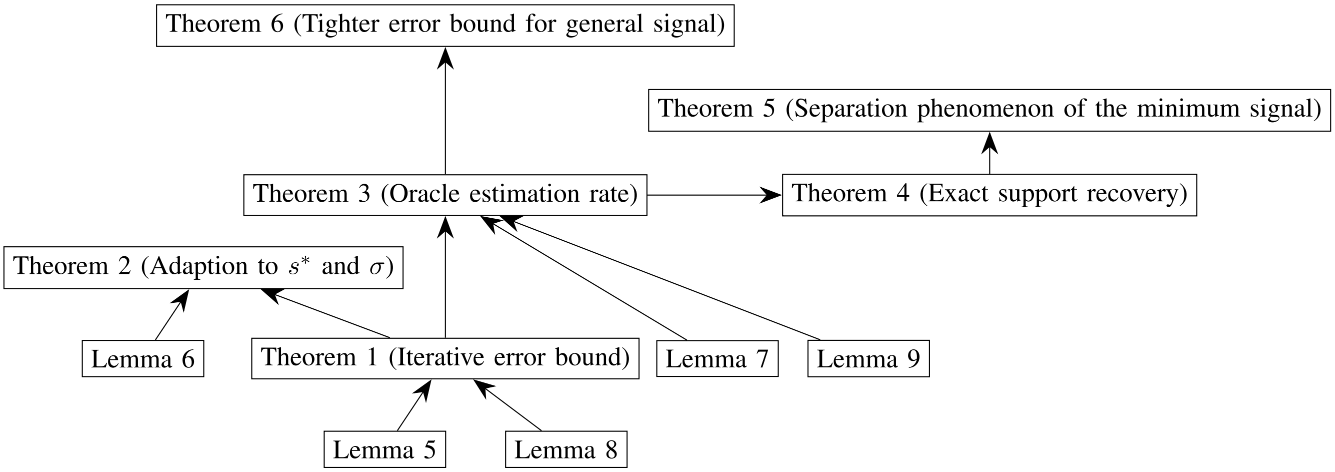

Fig. 8. A road map of the proof of the main results.

outperforms LASSO. These theoretical results inspire us to develop the sparsity parameter selection process.

Moreover, unlike the techniques based on complexity theory and empirical processes in convex algorithms such as LASSO [12], as well as other non-convex empirical risk-minimizing algorithms such as SCAD and MCP. References [13] and [14], our approach combines the analysis of iterative equations with a thorough understanding of the intrinsic properties of the hard thresholding estimator. We think this is the key aspect of our study that enables us to surpass the minimax optimality. To showcase the superiority of the nonconvex algorithm over its convex counterparts, we think that our approach merits significant attention and can be applied to other analyses, such as SCAD and MCP. In summary, the analysis of iterative algorithms can extend the properties of sequence models to practical methods, thereby yielding richer, more detailed, and significant conclusions on statistics.

Appendix
Roadmap of Proofs

Before presenting the proofs of the main theorems, we first provide a road map of the proofs (Figure 8).

Proof of Auxiliary Lemmas

Lemma 1: Assume $\boldsymbol{X}$ satisfies RIP$(3s, \delta_{3s})$. Then, for all $\forall S \subseteq [p]$ with size s, we have

$$\mathrm{pr}\left(\sum_{i\in S}\Xi_i^2 \le \frac{4\sigma^2 s\log(p/s)}{n}\right) \ge 1-\exp\left\{-Cs\log(p/s)\right\}$$

for some constant $C > 0$.

Proof 1: From Theorem 2.1 of [49], $\forall S \in \mathcal{S}^p(s)$, we have

$$\mathrm{pr}\left(\frac{\|\boldsymbol{X}_S^\top\boldsymbol{\xi}\|_2^2}{\sigma^2} \ge \mathrm{Tr}(\boldsymbol{X}_S^\top\boldsymbol{X}_S) + 2\|\boldsymbol{X}_S^\top\boldsymbol{X}_S\|_F\sqrt{l} + 2\lambda_{\max}(\boldsymbol{X}_S^\top\boldsymbol{X}_S)l\right) \le e^{-l}, \quad (14)$$

where constant $t \ge 0$. Since $\boldsymbol{X}_S \in \mathbb{R}^{n\times s}$ and $\|\boldsymbol{X}_j\|_2 = \sqrt{n}, j \in [p]$, we have

$$Tr(\boldsymbol{X}_S^\top\boldsymbol{X}_S) = \left(\sum_{j=1}^{s}\sum_{i=1}^{n}\boldsymbol{X}_{ij}^2\right) \le sn. \quad (15)$$

By RIP condition, we have

$$\lambda_{\max}(\boldsymbol{X}_S^\top\boldsymbol{X}_S) \le n(1+\delta). \quad (16)$$

On the other hand, from (15) and (16), we have

$$\|\boldsymbol{X}_S^\top\boldsymbol{X}_S\|_F = \sqrt{Tr(\boldsymbol{X}_S^\top\boldsymbol{X}_S\boldsymbol{X}_S^\top\boldsymbol{X}_S)} \le (1+\delta)\sqrt{sn}. \quad (17)$$

Substituting (15)–(17) into (14), we have

$$\mathrm{pr}\left(\frac{1}{n\sigma^2}\|\boldsymbol{X}_S^\top\boldsymbol{\xi}\|_2^2 \ge 2(1+\delta)\left(\sqrt{l}+\frac{\sqrt{s}}{2}\right)^2 + \frac{1-\delta}{2}s\right) \le e^{-l}.$$

Let $l = (1+C)s\log(p/s)$ for some small positive constant C. Since $\delta < 1$ and $\log(p/s) \gg 1$, we have $2(1+\delta)\left(\sqrt{l}+\frac{\sqrt{s}}{2}\right)^2 + \frac{1-\delta}{2}s < 4\ s\log(p/s)$. Consequently, we have

$$\mathrm{pr}\left(\frac{1}{n}\|\boldsymbol{X}_S^\top\boldsymbol{\xi}\|_2^2 \ge 4\sigma^2 s\log(p/s)\right) \le e^{-(1+C)s\log(p/s)}. \quad (18)$$

Note that

$$|\mathcal{S}^p(s)| \le \binom{p}{s} \le \left(\frac{ep}{s}\right)^s \le e^{s\log(ep/s)}. \quad (19)$$

Therefore, combining (18) and (19), we have

$$\begin{aligned}
&\mathrm{pr}\left(\forall S \in \mathcal{S}^p(s), \sum_{i\in S}\Xi_i^2 \le \frac{4\sigma^2 s\log(p/s)}{n}\right)\\
&= 1-\mathrm{pr}\left(\exists S \in \mathcal{S}^p(s), \sum_{i\in S}\Xi_i^2 > \frac{4\sigma^2 s\log(p/s)}{n}\right)\\
&\ge 1-|\mathcal{S}^p(s)|\mathrm{pr}\left(\sum_{i\in S}\Xi_i^2 > \frac{4\sigma^2 s\log(p/s)}{n}\right)\\
&\ge 1-e^{-Cs\log(p/s)},
\end{aligned}$$

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where the first inequality follows from the union bound. This completes the proof of Lemma 1.

Lemma 2: Assume that $\boldsymbol{\beta}^*$ is a s^*-sparse vector. Given a s-sparse vector $\boldsymbol{\beta}$, we have

$$\mathrm{pr}\left(\sup_{\|\boldsymbol{\beta}\|_0=s}\left\langle \boldsymbol{\xi}, \frac{\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)}{\|\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)\|_2}\right\rangle^2 \geq 4\sigma^2(s^*+s)\log\frac{p}{s^*+s}\right)$$
$$\leq e^{-s^*\log(p/s^*)}.$$

Proof 2: It holds that

$$\begin{aligned}
&\left|\left\langle \boldsymbol{\xi}, \frac{\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)}{\|\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)\|_2}\right\rangle\right| \\
=&\left|\left\langle \pi\boldsymbol{\xi}+(\mathbb{I}_n-\pi)\boldsymbol{\xi}, \frac{\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)}{\|\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)\|_2}\right\rangle\right| \\
=&\left|\left\langle \pi\boldsymbol{\xi}, \frac{\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)}{\|\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)\|_2}\right\rangle\right| \\
\leq&\|\pi\boldsymbol{\xi}\|_2,
\end{aligned}\tag{20}$$

where the first inequality follows from the property of the orthogonal projector, and the last inequality uses Cauchy-Schwartz inequality. Observe that $\boldsymbol{\beta}-\boldsymbol{\beta}^*$ has sparsity at most s^*+s. Hence,

$$\begin{aligned}
&\mathrm{pr}\left(\frac{\|\pi\boldsymbol{\xi}\|_2^2}{\sigma^2}\geq 4l\right) \\
\leq&\mathrm{pr}\left(\frac{\|\pi\boldsymbol{\xi}\|_2^2}{\sigma^2}\geq (s^*+s)+2\sqrt{(s^*+s)l}+2l\right) \\
\leq&e^{-l},
\end{aligned}\tag{21}$$

where the first inequality holds for $l \gg s^*+s$, and the last inequality follows from (14). Let $l=(s^*+s)\log\frac{p}{s^*+s}$, which satisfies $l \gg s^*+s$ for a sufficiently large p. From (20) and (21), we have

$$\mathrm{pr}\left(\left|\left\langle \boldsymbol{\xi}, \frac{\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)}{\|\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)\|_2}\right\rangle\right|^2 \geq 4\sigma^2(s^*+s)\log\frac{p}{s^*+s}\right)$$
$$\leq e^{-(s^*+s)\log\frac{p}{s^*+s}}.$$

Using union bound over all the subsets of $[p]$ with size s, we get

$$\begin{aligned}
&\mathrm{pr}\left(\sup_{\|\boldsymbol{\beta}\|_0=s}\left|\left\langle \boldsymbol{\xi}, \frac{\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)}{\|\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)\|_2}\right\rangle\right|^2 \geq 4\sigma^2(s^*+s)\log\frac{p}{s^*+s}\right) \\
\leq&\binom{p}{s}e^{-(s^*+s)\log\frac{p}{s}} \\
\leq&e^{-s^*\log(p/s^*)}.
\end{aligned}\tag{22}$$

Lemma 3: Assume that $\boldsymbol{\beta}^*$ is a s^*-sparse vector. Then, we have

$$\begin{aligned}
&\sup_{s\in\left[\frac{s^*}{\log(p/s^*)},4s^*\right]}\sup_{\|\boldsymbol{\beta}\|_0=s}\left|\left\langle \boldsymbol{\xi}, \frac{\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)}{\|\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)\|_2}\right\rangle\right|^2 \\
\leq&4\sigma^2\left(s^*+s\log\frac{p}{s}\right)
\end{aligned}$$

holds with probability at least $1-e^{-Cs^*}$.

Proof 3: Based on (21), for a fix $s>0$, we use a similar idea by letting $l=s^*+C_1 s\log\frac{p}{s}$, where $C_1\geq 1$ is a constant large enough, and $l\gg s^*+s$, then we obtain a similar inequality to (22):

$$\begin{aligned}
&\mathrm{pr}\left(\sup_{\|\boldsymbol{\beta}\|_0\leq s}\left|\left\langle \boldsymbol{\xi}, \frac{\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)}{\|\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)\|_2}\right\rangle\right|^2 \geq 4\sigma^2\left(s^*+C_1 s\log\frac{p}{s}\right)\right) \\
\leq&\binom{p}{s}e^{-C_0\left(s^*+C_1 s\log\frac{p}{s}\right)}.
\end{aligned}\tag{23}$$

For C_0, C_1 is large enough constant and $s\in\left[\frac{s^*}{\log(p/s^*)},4s^*\right]$, therefore we have

$$\mathrm{pr}\left(\sup_{\|\boldsymbol{\beta}\|_0\leq s}\left|\left\langle \boldsymbol{\xi}, \frac{\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)}{\|\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)\|_2}\right\rangle\right|^2 \geq g(s)\right)\leq e^{-Cg(s)},\tag{24}$$

where $g(s):=4\sigma^2\left(s^*+C_1 s\log\frac{p}{s}\right)\geq 8C_1 s^*$ for any $s\in\left[\frac{s^*}{\log(p/s^*)},4s^*\right]$.

The following step we use the peeling technique [33], [50] to extension (24) to any $s\in\left[\frac{s^*}{\log(p/s^*)},4s^*\right]$. Following the notations in Lemma 9 of [33], we let $\rho(u)=\|u\|_0$, $r=8C_1\sigma^2 s^*$, $a_n=1$, and

$$A:=\left\{\boldsymbol{\beta}\in\mathbb{R}^p:\|\boldsymbol{\beta}\|_0\in\left[\frac{s^*}{\log(p/s^*)},4s^*\right]\right\}.$$

Let

$$A_m:=\left\{\boldsymbol{\beta}\in A: 2^{m-1}r<g(\|\boldsymbol{\beta}\|_0)<2^m r\right\},\ m=1,2,\cdots,.$$

We define the random function:

$$f(\boldsymbol{\beta},\boldsymbol{X})=\left|\left\langle \boldsymbol{\xi}, \frac{\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)}{\|\boldsymbol{X}(\boldsymbol{\beta}-\boldsymbol{\beta}^*)\|_2}\right\rangle\right|^2,$$

and event $\mathcal{E}$,

$$\begin{aligned}
\mathcal{E}:=\Big\{&\exists\boldsymbol{\beta}\text{ such that }\|\boldsymbol{\beta}\|_0\in\left[\frac{s^*}{\log(p/s^*)},4s^*\right], \\
&f(\boldsymbol{\beta},\boldsymbol{X})>4\sigma^2\left(s^*+C_1\|\boldsymbol{\beta}\|_0\log\frac{p}{\|\boldsymbol{\beta}\|_0}\right)\Big\}.
\end{aligned}$$

Note that

$$\begin{aligned}
\mathrm{pr}(\mathcal{E})&\leq\sum_{m=1}^{\infty}\mathrm{pr}\left\{\exists\boldsymbol{\beta}\in A_m\text{ such that }f(\boldsymbol{\beta},\boldsymbol{X})\geq 2\ g(\|\boldsymbol{\beta}\|_0)\right\} \\
&\leq\sum_{m=1}^{\infty}\mathrm{pr}\left\{\sup_{\|\boldsymbol{\beta}\|_0\leq g^{-1}(2^m\mu)}f(\boldsymbol{\beta},\boldsymbol{X})\geq 2^m\mu\right\} \\
&\leq\sum_{m=1}^{\infty}2\exp\left(-ca_n\left[g\left(g^{-1}\left(2^m\mu\right)\right)\right]\right) \\
&=2\sum_{m=1}^{\infty}\exp\left(-ca_n 2^m\mu\right).
\end{aligned}$$

Consequently, we have $\mathrm{pr}(\mathcal{E})\leq C_1\exp(-C_2 s^*)$, which completes the proof of Lemma 3.

Lemma 4: Assume $\boldsymbol{X}$ satisfies RIP$(3s,\delta_{3s})$. Then, with probability at least $1-\exp\{-Cs^*\log(p/s^*)\}$, we have

$$\begin{aligned}
\|\boldsymbol{\beta}^*-\boldsymbol{\beta}^t\|_2\leq&\left(\frac{1+\delta_{3s}}{1-\delta_{3s}}\right)^{1/2}\|\boldsymbol{\beta}^*-\tilde{\boldsymbol{\beta}}^t\|_2+ \\
&4\sigma\left\{\frac{s+s^*}{(1-\delta_{3s})n}\log\frac{p}{s+s^*}\right\}^{1/2}.
\end{aligned}\tag{25}$$

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Proof 4: According to the definition of $\boldsymbol{\beta}^t$, we have

$$\|y - \boldsymbol{X}\boldsymbol{\beta}^t\|_2 \le \|y - \boldsymbol{X}\tilde{\boldsymbol{\beta}}^t\|_2.$$

Then, by some simple algebras, we have

$$\begin{aligned}&\|\boldsymbol{X}(\boldsymbol{\beta}^* - \boldsymbol{\beta}^t)\|_2^2 - 2|\langle \boldsymbol{\xi}, \boldsymbol{X}(\boldsymbol{\beta}^* - \boldsymbol{\beta}^t)\rangle| \\ \le& \|\boldsymbol{X}(\boldsymbol{\beta}^* - \tilde{\boldsymbol{\beta}}^t)\|_2^2 + 2|\langle \boldsymbol{\xi}, \boldsymbol{X}(\boldsymbol{\beta}^* - \tilde{\boldsymbol{\beta}}^t)\rangle|.\end{aligned}$$

From Lemma 2, with high probability, we obtain

$$\begin{aligned}&\|\boldsymbol{X}(\boldsymbol{\beta}^* - \boldsymbol{\beta}^t)\|_2^2 \\ \le& \|\boldsymbol{X}(\boldsymbol{\beta}^* - \tilde{\boldsymbol{\beta}}^t)\|_2^2 + 4\sigma\left\{(s+s^*)\log\frac{p}{s+s^*}\right\}^{1/2}.\end{aligned}$$

Using the RIP condition, we derive result (25).

Lemma 5: For any $\epsilon > 0$, there exists a constant $C_\epsilon > 0$ only depending on ϵ, such that we have

$$\mathrm{pr}\left(\sum_{i=1}^{p} \boldsymbol{\Xi}_i^2 \mathrm{I}\left\{|\boldsymbol{\Xi}_i| \ge \lambda_\infty^\epsilon\right\} \ge \frac{\sigma^2 s^*}{n\log(p/s^*)}\right) < \left(\frac{s^*}{p}\right)^{C_\epsilon}, \quad (26)$$

$$\mathrm{pr}\left(\sum_{i=1}^{p} \mathrm{I}\left\{|\boldsymbol{\Xi}_i| \ge \lambda_\infty^\epsilon\right\} \ge \frac{s^*}{\log(p/s^*)}\right) < \left(\frac{s^*}{p}\right)^{C_\epsilon}, \quad (27)$$

and

$$\mathrm{pr}\left(\sum_{i\in S^*} \mathrm{I}\left\{|\boldsymbol{\Xi}_i| \ge \frac{\epsilon\lambda_\infty^\epsilon}{2}\right\} \ge \frac{s^*}{\log(p/s^*)}\right) < \left(\frac{s^*}{p}\right)^{C_\epsilon}. \quad (28)$$

Proof 5: We first prove (26). From Lemma 4 of [31], we have for any $i \in [p]$,

$$\begin{aligned}&\mathbf{E}(\boldsymbol{\Xi}_i^2 \mathrm{I}\left\{|\boldsymbol{\Xi}_i| \ge \lambda_\infty^\epsilon\right\}) \\ \le& 2\sigma^2\left((1+\epsilon)^2\frac{\log(p/s^*)}{n} + \frac{2}{n}\right)e^{-(1+\epsilon)^2\log(p/s^*)}.\end{aligned}$$

By Markov's inequality, we have

$$\begin{aligned}&\mathrm{pr}\left(\sum_{i=1}^{p} \boldsymbol{\Xi}_i^2 \mathrm{I}\left\{|\boldsymbol{\Xi}_i| \ge \lambda_\infty^\epsilon\right\} \ge \frac{\sigma^2 s^*}{n\log(p/s^*)}\right) \\ \le& \frac{n\log(p/s^*)}{\sigma^2 s^*}\sum_{i=1}^{p} \mathbf{E}\left(\boldsymbol{\Xi}_i^2 \mathrm{I}\left\{|\boldsymbol{\Xi}_i| \ge \lambda_\infty^\epsilon\right\}\right) \\ \le& 4(1+\epsilon)^2(\log(p/s^*))^2\left(\frac{ep}{s^*}\right)^{-\epsilon}.\end{aligned}$$

Next, we turn to the proof of (27).

$$\begin{aligned}&\mathrm{pr}\left(\sum_{i=1}^{p} \mathrm{I}\left\{|\boldsymbol{\Xi}_i| \ge \lambda_\infty^\epsilon\right\} \ge \frac{s^*}{\log(p/s^*)}\right) \\ \le& \frac{\log(p/s^*)}{s^*}\sum_{i=1}^{p} \mathbf{E}\left(\mathrm{I}\left\{|\boldsymbol{\Xi}_i| \ge \lambda_\infty^\epsilon\right\}\right) \\ \le& \frac{p}{s^*}\log(p/s^*)\cdot 2e^{-(1+\epsilon)\log(p/s^*)} \\ \le& 4\log(p/s^*)\left(\frac{ep}{s^*}\right)^{-\epsilon},\end{aligned}$$

where the first inequality follows from Markov's inequality, and the second inequality uses the tail bound of sub-Gaussian distribution.

Finally we prove (28) by using Markov's inequality again:

$$\begin{aligned}&\mathrm{pr}\left(\sum_{i\in S^*} \mathrm{I}\left\{|\boldsymbol{\Xi}_i| \ge \frac{\epsilon\lambda_\infty^\epsilon}{2}\right\} \ge \frac{s^*}{\log(p/s^*)}\right) \\ \le& \frac{\log(p/s^*)}{s^*}\sum_{i\in S^*} \mathbf{E}\left(\mathrm{I}\left\{|\boldsymbol{\Xi}_i| \ge \frac{\epsilon\lambda_\infty^\epsilon}{2}\right\}\right) \\ \le& 2\log(p/s^*)\cdot\exp\left(-\frac{\epsilon^2(1+\epsilon)^2\log(p/s^*)}{4}\right) \\ \le& 2\log(p/s^*)\left(\frac{p}{s^*}\right)^{-\frac{\epsilon^2(1+\epsilon)^2}{4}}.\end{aligned}$$

Proof of Main Results

To simplify the notations, denote $\boldsymbol{H}^{t+1} := \boldsymbol{\beta}^t + \boldsymbol{X}^\top(y - \boldsymbol{X}\boldsymbol{\beta}^t)/n$. Denote $\lambda_\infty^\epsilon := (1+\epsilon)\sigma\{2\log(p/s^*)/n\}^{1/2}$ and $\tilde{\lambda}_\infty^\epsilon := (1+\epsilon)\sigma\{2\log p/n\}^{1/2}$. Denote Type-I and Type-II errors set in step t:

$$\begin{aligned}S_1^t &= S^t \cap (S^*)^c := \{i \in (S^*)^c : |\boldsymbol{H}_i^t| \ge \lambda_{s,t}\}, \\ S_2^t &= (S^t)^c \cap S^* := \{i \in S^* : |\boldsymbol{H}_i^t| < \lambda_{s,t}\},\end{aligned}$$

and

$$S_3^t := S^t \cap S^*.$$

Let $\langle \boldsymbol{x}, \boldsymbol{y}\rangle$ denote the dot product of vectors x and y.

Additionally, we introduce a scaling technique based on Assumption 2, which is frequently used in the following proofs. Recall that $\boldsymbol{H}^{t+1} = \boldsymbol{\beta}^* + \boldsymbol{\Phi}(\boldsymbol{\beta}^* - \boldsymbol{\beta}^t) + \boldsymbol{\Xi}$ and $\boldsymbol{\Phi} = \boldsymbol{X}^\top\boldsymbol{X}/n - \mathbb{I}_p$, $\boldsymbol{\Xi} = \boldsymbol{X}^\top\boldsymbol{\xi}/n$. For any subset S with $|S| = s$, if $\boldsymbol{\beta}^* - \boldsymbol{\beta}^t$ is $2s$-sparse, we have

$$\begin{aligned}&\sum_{i\in S}\langle \boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle^2 \\ \le& \sum_{i\in S'}\langle \boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle^2 \\ =& (\boldsymbol{\beta}^* - \boldsymbol{\beta}^t)_{S'}^\top \boldsymbol{\Phi}_{S'S'}^\top \boldsymbol{\Phi}_{S'S'}(\boldsymbol{\beta}^* - \boldsymbol{\beta}^t)_{S'} \\ \le& \|\boldsymbol{\Phi}_{S'S'}\|_2^2\left\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\right\|_2^2 \\ =& \left\{\left(\Lambda_{\max}\left(\frac{\boldsymbol{X}_{S'}^\top\boldsymbol{X}_{S'}}{n}\right) - 1\right) \vee \left(1 - \Lambda_{\min}\left(\frac{\boldsymbol{X}_{S'}^\top\boldsymbol{X}_{S'}}{n}\right)\right)\right\}^2 \\ &\times \left\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\right\|_2^2 \\ \le& \delta_{3s}^2\left\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\right\|_2^2,\end{aligned} \quad (29)$$

where S' is the union of S and the support set of $\boldsymbol{\beta} - \boldsymbol{\beta}^t$ and thus $|S'| \le 3s$.

A. Proof of Theorem 1

Proof 6: We prove Theorem 1 by analyzing the size of S_1^{t+1}, that is, the number of falsely discovered variables. First, we prove that

$$\|\boldsymbol{\beta}^{t+1} - \boldsymbol{\beta}^*\|_2 \le \gamma_{3s}\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2 + 12\sigma\left\{\frac{s\log(p/s)}{n}\right\}^{1/2} \quad (30)$$

holds for all $t \ge 0$. In particular, for $\forall t \ge 0$, we have $|S^{t+1}| = s \ge s^* = |S^*|$, which implies that $s \ge |S_1^{t+1}| \ge |S_2^{t+1}| \ge 0$. By the procedure of iterative hard thresholding operator, with probability greater than $1 - C_1\exp\{-C_2 s\log(p/s)\}$, we have

$$\sqrt{|S_1^{t+1}|\lambda_{s,t+1}^2} \le \sqrt{\sum_{i\in S_1^{t+1}}\left(\boldsymbol{H}_i^{t+1}\right)^2}$$

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$$\leq \sqrt{\sum_{i\in S_1^{t+1}} \langle \mathbf{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t \rangle^2} + \|\mathbf{\Xi}_{S_1^{t+1}}\|_2$$
$$\leq \delta_{3s}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2 + 2\sigma \left\{\frac{s\log(p/s)}{n}\right\}^{1/2}, \quad (31)$$

where the last inequality follows from (29) and Lemma 1.

For $\forall i \in [p]$, we have

$$\tilde{\boldsymbol{\beta}}_i^{t+1} - \boldsymbol{\beta}_i^* = -\boldsymbol{H}_i^{t+1}\mathrm{I}\{|\boldsymbol{H}_i^{t+1}| < \lambda_{s,t+1}\} + \left\langle \mathbf{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t \right\rangle + \Xi_i. \quad (32)$$

Therefore, with probability greater than $1 - C_1 \exp\{-C_2 s\log(p/s)\}$, we derive that

$$\|\tilde{\boldsymbol{\beta}}^{t+1} - \boldsymbol{\beta}^*\|_2$$
$$\leq \sqrt{\sum_{i\in S_2^{t+1}} \left(\boldsymbol{H}_i^{t+1}\right)^2 \mathrm{I}\{|\boldsymbol{H}_i^{t+1}| < \lambda_{s,t+1}\}}$$
$$+ \sqrt{\sum_{i\in S^*\cup S_1^{t+1}} \langle \mathbf{\Phi}_i, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle^2} + \sqrt{\sum_{i\in S^*\cup S_1^{t+1}} \Xi_i^2}$$
$$\overset{(i)}{\leq} \sqrt{|S_1^{t+1}|}\lambda_{s,t+1} + \delta_{3s}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2 + \sigma\left\{\frac{8s\log(p/s)}{n}\right\}^{1/2}$$
$$\overset{(ii)}{\leq} 2\delta_{3s}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2 + \left(2 + 2\sqrt{2}\right)\sigma\left\{\frac{s\log(p/s)}{n}\right\}^{1/2}, \quad (33)$$

where inequality (i) follows from $|S_2^{t+1}| \leq |S_1^{t+1}|$ and Lemma 1, and inequality (ii) follows from (31). Combining (33) and Lemma 4, we have

$$\|\boldsymbol{\beta}^{t+1} - \boldsymbol{\beta}^*\|_2 \leq 2\delta_{3s}\left(\frac{1+\delta_{3s}}{1-\delta_{3s}}\right)^{1/2}\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2 + 12\sigma\left\{\frac{s\log(p/s)}{n}\right\}^{1/2}.$$

Hence we obtain (30). By some simple calculations and $\boldsymbol{\beta}^0 = 0$, with high probability, we have

$$\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2 \leq \gamma_{3s}^t\|\boldsymbol{\beta}^*\|_2 + \frac{12\sigma}{1-\gamma_{3s}}\left\{\frac{s\log(p/s)}{n}\right\}^{1/2}, \forall t \geq 0.$$

which completes the proof of Theorem 1.

B. Proof of Theorem 2

Proof 7: We divide the proof of Theorem 2 into two steps. First, we prove (5), which illustrates that the selected model size is upper bounded by $2s^*$. Then, based on (5), we prove (6), that is, the estimator is minimax optimal. First, we introduce two penalties that are useful in the proof:

$$pen(s) := K\frac{s}{n}\log\left(\frac{p}{s}\right),$$
$$pen'(s) := -1 + \exp\left\{K\frac{s}{n}\log\left(\frac{p}{s}\right)\right\},$$

where K is a sufficiently large constant to be chosen later. For the simplicity of notation, denote $\|\boldsymbol{\xi}\|_n^2 = \sum_{i=1}^n \xi_i^2/n$ for all $\boldsymbol{\xi} \in \mathbb{R}^n$. Recall $\hat{\boldsymbol{\beta}}^{(s)}$ denotes the estimator obtained by Algorithm 1 given model size s. Define the optimal model size as:

$$\hat{s} := \arg\min_{s\in[s_{\max}]} \|y - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{(s)}\|_n^2\left(1 + pen'(s)\right). \quad (34)$$

Note that $1+pen'(s) \geq 1$ holds for all $s \in [s_{\max}]$. The definition of $\hat{s}$ yields that

$$\|y - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{(\hat{s})}\|_n^2 \leq \|y - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{(\hat{s})}\|_n^2(1 + pen'(\hat{s}))$$
$$\leq \|y - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{(s^*)}\|_n^2(1 + pen'(s^*)). \quad (35)$$

Proof of (5):

We prove (5) by contradiction. We assume that $\hat{s} > 2s^*$ holds at first. By the definitions of $pen'(s)$, we have

$$\frac{1 + pen'(\hat{s})}{1 + pen'(s^*)} = \exp\left\{\frac{K}{n}\hat{s}\log\frac{p}{\hat{s}} - \frac{K}{n}s^*\log\frac{p}{s^*}\right\}$$
$$\geq \exp\left\{\frac{K}{n}\frac{\hat{s}}{3}\log\frac{p}{\hat{s}}\right\}, \quad (36)$$

where the inequality follows from $s^*\log(p/s^*) \leq \frac{2}{3}\hat{s}\log(p/\hat{s})$ due to the assumption $\hat{s} > 2s^*$ and $p/s^* \geq 16$.

On one hand, with probability greater than $1-C\exp\left(-cs^*\log(p/s^*)\right)$, we have

$$\|y - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{(s^*)}\|_n^2$$
$$= \left\|\boldsymbol{X}\left(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(s^*)}\right)\right\|_n^2 + \|\boldsymbol{\xi}\|_n^2 - 2\langle\boldsymbol{\xi}, \frac{1}{n}\boldsymbol{X}(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(s^*)})\rangle$$
$$\leq \left\|\boldsymbol{X}\left(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(s^*)}\right)\right\|_n^2 + \|\boldsymbol{\xi}\|_n^2 + 2\left|\langle\boldsymbol{\xi}, \frac{1}{n}\boldsymbol{X}(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(s^*)})\rangle\right|$$
$$\leq \|\boldsymbol{\xi}\|_n^2 + (C_1 + C_2)\cdot\frac{\sigma^2 pen(s^*)}{K}, \quad (37)$$

where the last inequality follows from Corollary 1 and $RIP(3s^*)$ assumption, i.e.,

$$\mathrm{pr}\Bigg\{\left\|\boldsymbol{X}\left(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(s^*)}\right)\right\|_n^2 \leq (1+\delta_{3s^*})\left\|\hat{\boldsymbol{\beta}}^{(s^*)} - \boldsymbol{\beta}^*\right\|_2^2$$
$$\leq \underbrace{(1+\delta_{3s^*})\left(\frac{12}{1-\gamma_{3s^*}}\right)^2}_{=:C_1}\cdot\frac{\sigma^2 pen(s^*)}{K}\Bigg\}$$
$$\geq 1 - O\left(e^{-cs^*\log(p/s^*)}\right), \quad (38)$$

and Lemma 2, i.e.,

$$\mathrm{pr}\Bigg\{\left\langle\boldsymbol{\xi}, \boldsymbol{X}\left(\hat{\boldsymbol{\beta}}^{(s^*)} - \boldsymbol{\beta}^*\right)\right\rangle^2$$
$$\leq 8\sigma^2 s^*\log\left(\frac{p}{2s^*}\right)\left\|\boldsymbol{X}\left(\hat{\boldsymbol{\beta}}^{(s^*)} - \boldsymbol{\beta}^*\right)\right\|_2^2$$
$$\leq \underbrace{8C_1}_{=:C_2^2/4}\cdot n^2\left(\frac{\sigma^2 pen(s^*)}{K}\right)^2\Bigg\}$$
$$\geq 1 - O\left(e^{-cs^*\log(p/s^*)}\right),$$

where $c \in (0,1)$ can be chosen as an absolute constant.

For ease of display, in (39), we use the abbreviation $\hat{\Delta} := \boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(\hat{s})}$. Similarly, with probability greater than $1-C\exp\left(-cs^*\log(p/s^*)\right)$, we have

$$\|y - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{(\hat{s})}\|_n^2$$
$$= \|\boldsymbol{\xi}\|_n^2 + \left\|\boldsymbol{X}\left(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(\hat{s})}\right)\right\|_n^2 + \frac{2}{n}\left\langle\boldsymbol{\xi}, \boldsymbol{X}(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(\hat{s})})\right\rangle$$

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$$\geq \|\boldsymbol{\xi}\|_n^2 + \frac{1}{n}\|\boldsymbol{X}\hat{\Delta}\|_2^2 - \frac{2}{\sqrt{n}}\left|\left\langle \boldsymbol{\xi}, \frac{\boldsymbol{X}\hat{\Delta}}{\|\boldsymbol{X}\hat{\Delta}\|_2}\right\rangle\right| \cdot \frac{\|\boldsymbol{X}\hat{\Delta}\|_2}{\sqrt{n}}$$
$$\overset{(i)}{\geq} \|\boldsymbol{\xi}\|_n^2 - \frac{1}{n}\left\langle \boldsymbol{\xi}, \frac{\boldsymbol{X}\hat{\Delta}}{\|\boldsymbol{X}\hat{\Delta}\|_2}\right\rangle^2$$
$$\geq \|\boldsymbol{\xi}\|_n^2 - \underbrace{6}_{=:C_3} \cdot \frac{\sigma^2 pen(\hat{s})}{K}, \tag{39}$$

where inequality (i) follows from $xy \leq \frac{x^2}{4} + y^2$ for every $(x, y) \in \mathbb{R}^2$, and the last inequality follows from Lemma 2 again, i.e.,

$$\mathrm{pr}\left\{\left\langle \boldsymbol{\xi}, \frac{\boldsymbol{X}\hat{\Delta}}{\|\boldsymbol{X}\hat{\Delta}\|_2}\right\rangle^2 \leq 4\sigma^2(s^* + \hat{s})\log\left(\frac{p}{s^* + \hat{s}}\right) \leq 6n \cdot \frac{\sigma^2 pen(\hat{s})}{K}\right\}$$
$$\geq 1 - O\left(e^{-cs^*\log(p/s^*)}\right).$$

Also, in the derivation of (39) (especially the inequality (i)), we use the AM-GM inequality instead of any RIP-type assumption. This technique relaxes the assumption on $\mathbf{X}$, as we have discussed in Remark 2.

Therefore, under the assumption $\hat{s} \geq 2s^*$, the following results hold with probability greater than $1 - C\exp\left(-cs^*\log(p/s^*)\right)$:

$$\|y - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{(\hat{s})}\|_n^2 \cdot (1 + pen'(\hat{s}))$$
$$\overset{(34)}{\leq} \|y - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{(s^*)}\|_n^2 \cdot (1 + pen'(s^*))$$
$$\overset{(37),(39)}{\Longrightarrow} \left\{\|\boldsymbol{\xi}\|_n^2 - C_3 \cdot \frac{\sigma^2 pen(\hat{s})}{K}\right\} \cdot \frac{(1 + pen'(\hat{s}))}{(1 + pen'(s^*))}$$
$$\leq \|\boldsymbol{\xi}\|_n^2 + (C_1 + C_2) \cdot \frac{\sigma^2 pen(s^*)}{K}$$
$$\overset{(36)}{\Longrightarrow} \left\{\|\boldsymbol{\xi}\|_n^2 - C_3 \cdot \frac{\sigma^2 pen(\hat{s})}{K}\right\} \cdot \exp\left(\frac{1}{3}pen(\hat{s})\right)$$
$$\leq \|\boldsymbol{\xi}\|_n^2 + (C_1 + C_2) \cdot \frac{2\sigma^2 pen(\hat{s})}{3K}. \tag{40}$$

We next establish that the last line in (40) cannot hold with a high probability. Some preliminary concepts are defined below. By Hanson-Wright inequality [51], we have

$$\mathrm{pr}\left\{C_5\sigma^2 \leq \|\boldsymbol{\xi}\|_n^2 \leq C_6\sigma^2\right\} \geq 1 - O\left(\exp(-C_4 n)\right)$$
$$\geq 1 - O\left(e^{-cs^*\log(p/s^*)}\right), \tag{41}$$

where $C_5, C_6 > 0$ are two positive constants, and the last inequality follows from Assumption 3. Define

$$A_1 := \frac{3C_3}{C_5}, \quad A_2 := \frac{2(C_1 + C_2)}{C_5}, \quad K := 2A_1 + 2A_2,$$
$$\text{and} \quad x := \frac{1}{3K}pen(\hat{s}).$$

Additionally, we define a function

$$F(x) := (1 - A_1 x) \cdot \exp(Kx) - 1 - A_2\, x, \quad x > 0.$$

Assumption 3 leads that $x \leq \frac{s_{\max}\log(p/s_{\max})}{3n} \leq 1/K$ by taking $C_0 = K/3$. Therefore, to prove the contradiction, we need to prove that $F(x) > 0$ holds for every $x \in (0, 1/K)$. Note that $F(0) = 0$ and

$$F'(x) = \exp(Kx)(K - A_1 - A_1 Kx) - A_2$$
$$\geq K - 2A_1 - A_2 = A_2 > 0, \quad \text{for every } x \in \left(0, \frac{1}{K}\right),$$

which leads $F(x) > 0$ for every $x \in (0, 1/K)$, and proves that

$$\mathrm{pr}\left\{\left(\|\boldsymbol{\xi}\|_n^2 - C_3 \cdot \frac{\sigma^2 pen(\hat{s})}{K}\right) \cdot \exp\left(\frac{1}{3}pen(\hat{s})\right) > \|\boldsymbol{\xi}\|_n^2 + (C_1 + C_2) \cdot \frac{2\sigma^2 pen(\hat{s})}{3K}\right\}$$
$$\geq 1 - O\left(e^{-cs^*\log(p/s^*)}\right),$$

where the probability follows from (41). This result yields a contradiction and proves that

$$\mathrm{pr}\{\hat{s} \leq 2s^*\} \geq 1 - O\left(e^{-cs^*\log(p/s^*)}\right),$$

which completes the first part of Theorem 2.

Proof of (6):

By the definition of $\hat{s}$, with probability greater than $1 - O\left(e^{-cs^*\log(p/s^*)}\right)$ we derive that

$$\left\|y - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{(\hat{s})}\right\|_2^2 (1 + pen'(\hat{s})) \leq \|y - \boldsymbol{X}\hat{\boldsymbol{\beta}}^{(s^*)}\|_2^2(1 + pen'(s^*))$$
$$\Longrightarrow \left\|\boldsymbol{X}\left(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(\hat{s})}\right)\right\|_2^2 + \|\boldsymbol{\xi}\|_2^2 + 2\langle \boldsymbol{\xi}, \boldsymbol{X}(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(\hat{s})})\rangle$$
$$\leq \left\{\left\|\boldsymbol{X}\left(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(s^*)}\right)\right\|_2^2 + \|\boldsymbol{\xi}\|_2^2 + 2\left|\langle \boldsymbol{\xi}, \boldsymbol{X}(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(s^*)})\rangle\right|\right\}$$
$$\times \left(1 + pen'(s^*)\right)$$
$$\Longrightarrow \left\|\boldsymbol{X}\left(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(\hat{s})}\right)\right\|_2^2 - 2\left|\langle \boldsymbol{\xi}, \boldsymbol{X}(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(\hat{s})})\rangle\right|$$
$$\leq \left\{\left\|\boldsymbol{X}\left(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(s^*)}\right)\right\|_2^2 + 2\left|\langle \boldsymbol{\xi}, \boldsymbol{X}(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(s^*)})\rangle\right|\right\} \times$$
$$\left(1 + pen'(s^*)\right) + \|\boldsymbol{\xi}\|_2^2 \cdot pen'(s^*)$$
$$\overset{(i)}{\Longrightarrow} \left\|\boldsymbol{X}\left(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(\hat{s})}\right)\right\|_2^2 - 2\left|\langle \boldsymbol{\xi}, \boldsymbol{X}(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(\hat{s})})\rangle\right|$$
$$\leq \left\{\left\|\boldsymbol{X}\left(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(s^*)}\right)\right\|_2^2 + 2\left|\langle \boldsymbol{\xi}, \boldsymbol{X}(\boldsymbol{\beta}^* - \hat{\boldsymbol{\beta}}^{(s^*)})\rangle\right|\right\} \cdot e^3$$
$$+ C_6\left(\frac{e^3 - 1}{3}\right)K \cdot \sigma^2 s^*\log(p/s^*) \tag{42}$$

where (i) follows from Assumption 3 with taking $C_0 = K/3$ (leading $1 + pen'(s^*) = \exp\left\{\frac{Ks^*\log(p/s^*)}{n}\right\} \leq \exp(3)$ and $pen'(s^*) \leq \frac{e^3-1}{3}\frac{Ks^*\log(p/s^*)}{n}$) and (41).

Then, by combining the first result $\mathrm{pr}\{\hat{s} \leq 2s^*\} \geq 1 - O\left(e^{-cs^*\log(p/s^*)}\right)$ and Lemma 2, we establish the following inequalities with probability greater than $1 - O\left(e^{-cs^*\log(p/s^*)}\right)$:

$$\left\langle \boldsymbol{\xi}, \boldsymbol{X}(\hat{\boldsymbol{\beta}}^{(s^*)} - \boldsymbol{\beta}^*)\right\rangle^2 \leq \|\boldsymbol{X}(\hat{\boldsymbol{\beta}}^{(s^*)} - \boldsymbol{\beta}^*)\|_2^2 \cdot 8\sigma^2 s^*\log(p/s^*),$$
$$\left\langle \boldsymbol{\xi}, \boldsymbol{X}(\hat{\boldsymbol{\beta}}^{(\hat{s})} - \boldsymbol{\beta}^*)\right\rangle^2 \leq \|\boldsymbol{X}(\hat{\boldsymbol{\beta}}^{(\hat{s})} - \boldsymbol{\beta}^*)\|_2^2 \times 4\sigma^2(s^* + \hat{s})\log\left(\frac{p}{s^* + \hat{s}}\right)$$
$$\leq \|\boldsymbol{X}(\hat{\boldsymbol{\beta}}^{(\hat{s})} - \boldsymbol{\beta}^*)\|_2^2 \cdot 12\sigma^2 s^*\log(p/s^*). \tag{43}$$

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Therefore, combining (43) and (42), we establish the following inequalities with probability greater than $1 - O\left(e^{-cs^* \log(p/s^*)}\right)$:

$$\begin{aligned}
&\left\| X\left(\beta^* - \hat{\beta}^{(\hat{s})}\right)\right\|_2^2 - \frac{1}{2}\left\| X\left(\beta^* - \hat{\beta}^{(\hat{s})}\right)\right\|_2^2 - 24\sigma^2 s^* \log(p/s^*) \\
&\overset{(i)}{\le} \left\| X\left(\beta^* - \hat{\beta}^{(\hat{s})}\right)\right\|_2^2 - 2\left|\langle \xi, X(\beta^* - \hat{\beta}^{(\hat{s})})\rangle\right| \\
&\le e^3\left\{\left\| X\left(\beta^* - \hat{\beta}^{(s^*)}\right)\right\|_2^2 + 2\left|\langle \xi, X(\beta^* - \hat{\beta}^{(s^*)})\rangle\right|\right\} + C_6\left(\frac{e^3-1}{3}\right)K\sigma^2 s^* \log(p/s^*) \\
&\overset{(ii)}{\le} e^3\left\{\left\| X\left(\beta^* - \hat{\beta}^{(s^*)}\right)\right\|_2^2 + \frac{1}{2}\left\| X\left(\beta^* - \hat{\beta}^{(s^*)}\right)\right\|_2^2 + 16\sigma^2 s^* \log(p/s^*)\right\} + C_6\left(\frac{e^3-1}{3}\right)K\sigma^2 s^* \log(p/s^*) \\
&\le \left(\frac{3e^3 C_1}{2} + 16e^3 + C_6 K\frac{e^3-1}{3}\right)\sigma^2 s^* \log(p/s^*),
\end{aligned}$$

where both inequality (i) and (ii) follow from (43) and the AM-GM inequality $2xy \le \frac{x^2}{2} + 2y^2$ for every $(x, y) \in \mathbb{R}^2$. The last inequality follows from (38).

Finally, by simply using $RIP(3s^*, \delta_{3s^*})$ assumption, with probability greater than $1 - O\left(e^{-cs^* \log(p/s^*)}\right)$ we have

$$\begin{aligned}
\left\|\beta^* - \hat{\beta}^{(\hat{s})}\right\|_2^2 &\le \frac{\left\| X\left(\beta^* - \hat{\beta}^{(\hat{s})}\right)\right\|_2^2}{n(1-\delta_{3s^*})} \\
&\le \underbrace{\frac{2}{1-\delta_{3s^*}}\left(24 + \frac{3e^3C_1}{2} + 16e^3 + C_6K\frac{e^3-1}{3}\right)}_{=:C_7} \times \frac{\sigma^2 s^* \log(p/s^*)}{n},
\end{aligned}$$

which completes the proof of (6).

C. Proof of Theorem 3

Recall that $\lambda_\infty^\epsilon = (1+3\epsilon)\sigma\sqrt{\frac{2\log(p/s^*)}{n}}$. Here we consider two cases: $\lambda_{s,t+1} \ge \lambda_\infty^\epsilon$ and $\lambda_{s,t+1} < \lambda_\infty^\epsilon$, where we take $\lambda_{s,t+1}$ as the minimal absolute signal of the selected variables in the $(t+1)$-th iteration based on model size s.

It is quite intuitive that when $\lambda_{s,t+1} \ge \lambda_\infty^\epsilon$, we could use those falsely discovered signals (that is, the elements in S_1^{t+1}) to control those signals in S_2^{t+1}. On the other hand, when $\lambda_{s,t+1} < \lambda_\infty^\epsilon$, we could separate S_1^{t+1} into two parts:

$$S_{11}^{t+1} := \{i \in S_1^{t+1} : \lambda_{s,t+1} < |\Xi_i + \langle \Phi_i^\top, \beta^* - \beta^t\rangle| \le \lambda_\infty^\epsilon\} \quad (44)$$

and $S_1^{t+1} \setminus S_{11}^{t+1}$ as well. Then, we use S_2^{t+1} to control S_{11}^{t+1}, while signals in $S_1^{t+1} \setminus S_{11}^{t+1}$ could be controlled as the same as the first case. To simplify the proofs, we provide the following two lemmas.

Lemma 6: Assume all the conditions in Theorem 3 hold. For all $t \ge 0$ and all $\tau \in [0, 3s^*]$, we have

$$\|\tilde{\beta}^{t+1} - \beta^*\|_2 \le \left(1 + 2\sqrt{2} + \frac{3\sqrt{2}}{\epsilon}\right)\delta_{3s}\|\beta^t - \beta^*\|_2 + \left(4 + 2\epsilon + \frac{4}{\epsilon}\right)\sigma\left\{\frac{s^* \vee \left(\tau\log(p/s^*)\right)}{n}\right\}^{1/2}. \quad (45)$$

holds with probability at least $1 - \exp\left(-C_{1,\epsilon}s^*\right) - (s^*/p)^{C_{2,\epsilon}}$, where $C_{1,\epsilon}$ and $C_{2,\epsilon}$ are positive constants depending on ϵ.

Lemma 7: Assume all the conditions in Theorem 3 hold. With probability larger than $1 - (s^*/p)^{C_\epsilon}$, the type-I error satisfies that

$$|S_1^{t+1}| \le \frac{s^*}{\log(p/s^*)} + \frac{4\delta_{3s}^2\|\beta^t - \beta^*\|_2^2}{\epsilon^2\left(\lambda_\infty^0\right)^2} + \tau. \quad (46)$$

First, we assume that the above two lemmas 6 and 7 hold, and prove Theorem 3. Then we prove Lemma 6 and 7 respectively.

Proof 8 (Proof of Theorem 3): Note that $\|y - X\beta^{t+1}\|_2 \le \|y - X\tilde{\beta}^{t+1}\|_2$. Then, by some simple algebras, we have

$$\begin{aligned}
&\|X(\beta^* - \beta^{t+1})\|_2^2 - 2|\langle \xi, X(\beta^* - \beta^{t+1})\rangle| \\
\le &\|X(\beta^* - \tilde{\beta}^{t+1})\|_2^2 + 2|\langle \xi, X(\beta^* - \tilde{\beta}^{t+1})\rangle|. \quad (47)
\end{aligned}$$

Since the sparsity of $\beta^* - \beta^t$ is at most $s^* + |S_1^t|$. Similar to the proof of Lemma 2, we set $l = s^* + |S_1^t|\log p/s^*$ in (21). Consequently, from Lemma 3, with probability larger than $1 - \exp(-Cs^*)$, we have

$$\left\langle \xi, \frac{X(\beta^* - \beta^{t+1})}{\|X(\beta^* - \beta^{t+1})\|_2}\right\rangle^2 \le 4\sigma^2\left(s^* + |S_1^{t+1}|\log\left(\frac{p}{s^*}\right)\right).$$

Combining (46) in Lemma 7, we have

$$\begin{aligned}
&\left\langle \xi, \frac{X(\beta^* - \beta^{t+1})}{\|X(\beta^* - \beta^{t+1})\|_2}\right\rangle^2 \\
&\le 4\sigma^2\left(2s^* + \frac{4\delta_{3s}^2 n\|\beta^t - \beta^*\|_2^2}{\sigma^2\epsilon^2} + \tau\log p/s^*\right) \\
&\le 8\sigma^2 s^* + 16\, n\delta_{3s}\|\beta^t - \beta^*\|_2^2 + 4\sigma^2\tau\log p/s^*, \quad (48)
\end{aligned}$$

where the second inequality uses $\delta < \epsilon^2$. The term $|\langle \xi, X(\beta^* - \tilde{\beta}^t)\rangle|$ has a similar result as (48). Combining (47) and (48), we have

$$\begin{aligned}
&\|\beta^* - \beta^{t+1}\|_2 \\
&\le \left(\frac{1+\delta_{3s}}{1-\delta_{3s}}\right)^{1/2}\|\beta^* - \tilde{\beta}^{t+1}\|_2 + \left\{\frac{4\sigma^2}{(1-\delta_{3s})n}\left(8s^* + \frac{16\delta_{3s}n}{\sigma^2}\|\beta^t - \beta^*\|_2^2 + 4\tau\log p/s^*\right)\right\}^{1/2}. \quad (49)
\end{aligned}$$

From (45) and (49), with high probability, we have

$$\begin{aligned}
&\|\beta^{t+1} - \beta^*\|_2 \\
&\le \frac{\sqrt{1+\delta_{3s}}\left(1 + 2\sqrt{2} + \frac{3\sqrt{2}}{\epsilon}\right)\delta_{3s} + 8\sqrt{\delta_{3s}}}{\sqrt{1-\delta_{3s}}}\|\beta^t - \beta^*\|_2 + C_1\sigma\left\{\frac{s^* \vee \tau\log(p/s^*)}{n}\right\}^{1/2}. \\
&\le \widetilde{\gamma}_{3s,\epsilon}\|\beta^t - \beta^*\|_2 + C_1\sigma\left\{\frac{s^* \vee \tau\log(p/s^*)}{n}\right\}^{1/2},
\end{aligned}$$

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where the second inequality follows from $\widetilde{\gamma}_{3s,\epsilon} < 1$. By using the above inequality repeatedly, with high probability, we have

$$\begin{aligned}&\|\boldsymbol{\beta}^{t+1} - \boldsymbol{\beta}^*\|_2 \\ &\leq \widetilde{\gamma}^t_{3s,\epsilon}\|\boldsymbol{\beta}^0 - \boldsymbol{\beta}^*\|_2 + C_1\sigma\left\{\frac{s^* \vee \tau\log(p/s^*)}{n}\right\}^{1/2} \\ &\leq \widetilde{\gamma}^t_{3s,\epsilon}\sigma\left(\frac{s^*\log p/s^*}{n}\right)^{1/2} + C_1\sigma\left\{\frac{s^* \vee \tau\log(p/s^*)}{n}\right\}^{1/2}.\end{aligned}$$

On one hand, when $s^*/\log(p/s^*) < \tau \leq 3s^*$, that is, $s \in (s^* + s^*/\log(p/s^*),\ 4s^*]$, for $t \geq \log\left(C_2^2 s^*/\tau\right)$, with high probability, we have

$$\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2 \leq C_1\sigma\left\{\frac{\tau\log(p/s^*)}{n}\right\}^{1/2}.$$

On the other hand, when $0 \leq \tau \leq s^*/\log(p/s^*)$, that is, $s \in [s^*,\ s^* + s^*/\log(p/s^*)]$, for $t \geq \log\left(C_2^2\log p/s^*\right)$, with high probability, we have

$$\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2 \leq C_1\sigma\left(\frac{s^*}{n}\right)^{1/2}.$$

Thus we complete the proof of Theorem 3.

1) Proof of Lemma 6: Proof 9: Define $A_i^{t+1} := \Xi_i + \langle\boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle$ for all $i \in [p]$. Note that $|S_1^{t+1}| \geq |S_2^{t+1}|$ and for any $i \in S_1^{t+1}, i' \in S_2^{t+1}$,we have $|A_i^{t+1}| \geq \lambda_{s,t+1} > |\boldsymbol{\beta}^*_{i'} + \Xi_{i'} + \langle\boldsymbol{\Phi}_{i'}^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| = |\boldsymbol{H}_{i'}^{t+1}|$. Consequently,

$$\sqrt{\sum_{i'\in S_2^{t+1}}\left(\boldsymbol{H}_{i'}^{t+1}\right)^2 \mathrm{I}\{|\boldsymbol{H}_{i'}^{t+1}| < \lambda_{s,t+1}\}} \leq \sqrt{\sum_{i\in S_1^{t+1}}(A_i^{t+1})^2}. \quad (50)$$

We have

$$\begin{aligned}&\|\tilde{\boldsymbol{\beta}}^{t+1} - \boldsymbol{\beta}^*\|_2 \\ &= \Bigg\{\sum_{i\in S_1^{t+1}\cup S_3^{t+1}}(A_i^{t+1})^2 + \sum_{i\in S_2^{t+1}}\Xi_i^2 \\ &\quad + \langle\boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^t - \boldsymbol{\beta}^*\rangle^2 + \left(\boldsymbol{H}_i^{t+1}\right)^2\mathrm{I}\{|\boldsymbol{H}_i^{t+1}| < \lambda_{s,t+1}\}\Bigg\}^{1/2} \\ &\leq \|\langle\boldsymbol{\Phi}^\top_{S_2^{t+1}\cup S_3^{t+1}}, \boldsymbol{\beta}^t - \boldsymbol{\beta}^*\rangle\|_2 + \|\Xi_{S_2^{t+1}\cup S_3^{t+1}}\|_2 + \\ &\quad \sqrt{2\sum_{i\in S_1^{t+1}}(A_i^{t+1})^2}, \end{aligned} \quad (51)$$

where the first inequality follows from the definition of sets S_1, S_2, S_3 and triangle inequality.

Next, we divide the remaining proof into two cases.

Case 1: $\lambda_{s,t+1} \geq \lambda_\infty^\epsilon$. With probability greater than $1 - \left(\frac{s^*}{p}\right)^{C_1}$, we have

$$\begin{aligned}&\sqrt{\sum_{i\in S_1^{t+1}}(A_i^{t+1})^2} \\ &\leq \left(\sum_{i\in S_1^{t+1}}\Xi_i^2\mathrm{I}\left\{|\Xi_i + \langle\boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| \geq \lambda_\infty^\epsilon\right\}\right)^{1/2} + \\ &\quad \|\langle\boldsymbol{\Phi}^\top_{S_1^{t+1}}, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle\|_2 \\ &\leq \left(\sum_{i\in S_1^{t+1}}\Xi_i^2\mathrm{I}\left\{|\Xi_i| \geq \lambda_\infty^{\frac{\epsilon}{2}}\right\}\right)^{1/2} + \\ &\quad \left(\sum_{i\in S_1^{t+1}}\Xi_i^2\mathrm{I}\left\{|\Xi_i| \leq \lambda_\infty^{\frac{\epsilon}{2}}, |\langle\boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| \geq \frac{\epsilon\lambda_\infty^0}{2}\right\}\right)^{1/2} \\ &\quad + \|\langle\boldsymbol{\Phi}^\top_{S_1^{t+1}}, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle\|_2 \\ &\leq \sigma\sqrt{\frac{s^*}{n\log(p/s^*)}} + 2\left(1 + \frac{1}{\epsilon}\right)\delta_{3s}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2, \end{aligned} \quad (52)$$

where the last inequality follows from (26) of Lemma 5.

Next we will control $\|\Xi_{S^*}\|_2$. Similarly to Lemma 1, with the fixed support set S^* and $\forall t > 0$, we have

$$\mathrm{pr}\left(\frac{n}{\sigma^2}\|\Xi_{S^*}\|_2^2 \geq s^* + 2(1+\delta)\sqrt{s^* t} + 2(1+\delta)t\right) \leq e^{-t},$$

then, based on $t = \frac{s^*}{4}$ and $\delta < 1$, we obtain that

$$\mathrm{pr}\left(\|\Xi_{S^*}\|_2^2 \geq 4\sigma^2\frac{s^*}{n}\right) \leq \exp\left(-\frac{s^*}{4}\right). \quad (53)$$

Then, with probability greater than $1 - \left(\frac{s^*}{p}\right)^{C_1} - \exp(-C_2 s^*)$, we combine (51)–(53) and obtain

$$\begin{aligned}&\|\tilde{\boldsymbol{\beta}}^{t+1} - \boldsymbol{\beta}^*\|_2 \\ &\leq \|\langle\boldsymbol{\Phi}^\top_{S_2^{t+1}\cup S_3^{t+1}}, \boldsymbol{\beta}^t - \boldsymbol{\beta}^*\rangle\|_2 + \|\Xi_{S_2^{t+1}\cup S_3^{t+1}}\|_2 + \\ &\quad \sqrt{2\sum_{i\in S_1^{t+1}}(A_i^{t+1})^2} \\ &\leq \left(1 + 2\sqrt{2} + \frac{2\sqrt{2}}{\epsilon}\right)\delta_{3s}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2 + 2\sigma\left(\frac{s^*}{n}\right)^{1/2} + \\ &\quad \sigma\sqrt{\frac{2s^*}{n\log(p/s^*)}} \\ &\leq \left(1 + 2\sqrt{2} + \frac{2\sqrt{2}}{\epsilon}\right)\delta_{3s}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2 + 4\sigma\left(\frac{s^*}{n}\right)^{1/2}. \end{aligned} \quad (54)$$

Case 2: $\lambda_{s,t+1} < \lambda_\infty^\epsilon$. Consider $S_{11}^{t+1} := \{i \in S_1^{t+1} : |A_i^{t+1}| \leq \lambda_\infty^\epsilon\}$. Note that for any $j \in S_2^{t+1}$, we have

$$\begin{aligned}&|\boldsymbol{\beta}^*_j| - |\Xi_j + \langle\boldsymbol{\Phi}_j^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| \\ &\leq |\boldsymbol{\beta}^*_j + \Xi_j + \langle\boldsymbol{\Phi}_j^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| = |\boldsymbol{H}_j^{t+1}| < \lambda_{s,t+1} < \lambda_\infty^\epsilon,\end{aligned}$$

which leads

$$\begin{aligned}&|A_j^{t+1}| \\ &= |\Xi_j + \langle\boldsymbol{\Phi}_j^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| > \left|\boldsymbol{\beta}^*_j\right| - \lambda_\infty^\epsilon \geq 2\epsilon\lambda_\infty^0 \geq \epsilon\lambda_\infty^\epsilon.\end{aligned}$$

Since $|S_{11}^{t+1}| \leq |S_1^{t+1}| = |S_2^{t+1}| + \tau$, we have

$$\begin{aligned}\|A^{t+1}_{S_{11}^{t+1}}\|_2 &\leq \sqrt{\sum_{i\in S_{11}^{t+1}}\left(\lambda_\infty^\epsilon\right)^2} \\ &\leq \sqrt{\sum_{j\in S_2^{t+1}}\frac{\left(A_j^{t+1}\right)^2}{\epsilon^2}} + \sqrt{\tau\left(\lambda_\infty^\epsilon\right)^2} \\ &\leq \frac{1}{\epsilon}\left\|A^{t+1}_{S_2^{t+1}}\right\|_2 + (1+\epsilon)\sigma\left\{\frac{2\tau\log(p/s^*)}{n}\right\}^{1/2}\end{aligned}$$

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$$\leq \frac{1}{\epsilon}\left(\|\boldsymbol{\Xi}_{S^*}\|_2 + \delta_{3s}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2\right) + (1+\epsilon)\sigma\left\{\frac{2\tau\log(p/s^*)}{n}\right\}^{1/2}.$$

As a consequence, with high probability, we have

$$\begin{aligned}\|A^{t+1}_{S_1^{t+1}}\|_2 &\leq \|A^{t+1}_{S_{11}^{t+1}}\|_2 + \|A^{t+1}_{S_1^{t+1}\setminus S_{11}^{t+1}}\|_2 \\ &\leq \frac{2\sigma}{\epsilon}\left(\frac{s^*}{n}\right)^{1/2} + \frac{1}{\epsilon}\delta_{3s}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2 + (1+\epsilon)\sigma\left\{\frac{2\tau\log(p/s^*)}{n}\right\}^{1/2} \\ &\quad + \sigma\sqrt{\frac{s^*}{n\log(p/s^*)}} + 2\left(1+\frac{1}{\epsilon}\right)\delta_{3s}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2 \\ &< \left(2+\frac{3}{\epsilon}\right)\delta_{3s}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2 + \left(1+\frac{2}{\epsilon}\right)\sigma\left(\frac{s^*}{n}\right)^{1/2} \\ &\quad + (1+\epsilon)\sigma\left\{\frac{2\tau\log(p/s^*)}{n}\right\}^{1/2}. \end{aligned} \tag{55}$$

Similar to (54), with high probability, we combine (51), (53), (55) and obtain

$$\begin{aligned}&\|\tilde{\boldsymbol{\beta}}^{t+1} - \boldsymbol{\beta}^*\|_2 \\ &\leq \|\langle \boldsymbol{\Phi}^\top_{S_2^{t+1}\cup S_3^{t+1}}, \boldsymbol{\beta}^t - \boldsymbol{\beta}^*\rangle\|_2 + \|\boldsymbol{\Xi}_{S_2^{t+1}\cup S_3^{t+1}}\|_2 + \sqrt{2\sum_{i\in S_1^{t+1}}(A_i^{t+1})^2} \\ &\leq \left(1+2\sqrt{2}+\frac{3\sqrt{2}}{\epsilon}\right)\delta_{3s}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2 + \left(2+\sqrt{2}+\frac{2\sqrt{2}}{\epsilon}\right)\sigma\left(\frac{s^*}{n}\right)^{1/2} \\ &\quad + 2(1+\epsilon)\sigma\left\{\frac{\tau\log(p/s^*)}{n}\right\}^{1/2} \\ &< \left(1+2\sqrt{2}+\frac{3\sqrt{2}}{\epsilon}\right)\delta_{3s}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2 + \left(4+2\epsilon+\frac{4}{\epsilon}\right)\sigma\left\{\frac{s^*\vee\left(\tau\log(p/s^*)\right)}{n}\right\}^{1/2}.\end{aligned}$$

According to **case 1** and **case 2**, with high probability, we have

$$\|\tilde{\boldsymbol{\beta}}^{t+1} - \boldsymbol{\beta}^*\|_2 \leq \left(1+2\sqrt{2}+\frac{3\sqrt{2}}{\epsilon}\right)\delta_{3s}\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2 + \left(4+2\epsilon+\frac{4}{\epsilon}\right)\sigma\left\{\frac{s^*\vee\left(\tau\log(p/s^*)\right)}{n}\right\}^{1/2}. \tag{56}$$

which implies that (45) holds for all $t \geq 0$.

2) Proof of Lemma 7: Proof 10: Note that $s = s^* + \tau \in [s^*, 4s^*]$. We separate the analysis into two cases:

Case 1: if $\lambda_{s,t+1} \leq \lambda_\infty^\epsilon$, we have

$$\begin{aligned}&|S_1^{t+1}| - \tau = |S_2^{t+1}| \\ &\leq \left|\left\{i\in S^* : |\boldsymbol{\Xi}_i + \langle \boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| \geq |\boldsymbol{\beta}_i^*| - \lambda_{s,t+1}\right\}\right| \\ &\leq \left|\left\{i\in S^* : |\boldsymbol{\Xi}_i + \langle \boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| \geq |\boldsymbol{\beta}_i^*| - \lambda_\infty^\epsilon\right\}\right| \\ &\leq \left|\left\{i\in S^* : |\boldsymbol{\Xi}_i + \langle \boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| \geq \epsilon\lambda_\infty^\epsilon\right\}\right|.\end{aligned}$$

Then, with high probability, we have

$$\begin{aligned}|S_1^{t+1}| &\leq \sum_{i\in S^*}\mathrm{I}\left\{|\boldsymbol{\Xi}_i + \langle \boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| \geq \epsilon\lambda_\infty^\epsilon\right\} + \tau \\ &\leq \sum_{i\in S^*}\mathrm{I}\left\{|\boldsymbol{\Xi}_i| \geq \frac{\epsilon\lambda_\infty^\epsilon}{2}\right\} + \sum_{i\in S^*}\mathrm{I}\left\{|\langle \boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| \geq \frac{\epsilon\lambda_\infty^\epsilon}{2}\right\} + \tau \\ &\leq \frac{s^*}{\log(p/s^*)} + \frac{4\delta_{3s}^2\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2^2}{\epsilon^2\left(\lambda_\infty^\epsilon\right)^2} + \tau.\end{aligned} \tag{57}$$

where the third inequality follows from (28) of Lemma 5.

Case 2: if $\lambda_{s,t+1} > \lambda_\infty^\epsilon$, we have

$$\begin{aligned}&\tau + |S_2^{t+1}| = |S_1^{t+1}| \\ &\leq \left|\left\{i\in (S^*)^c : |\boldsymbol{\Xi}_i + \langle \boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| \geq \lambda_{s,t+1}\right\}\right| \\ &\leq \left|\left\{i\in (S^*)^c : |\boldsymbol{\Xi}_i + \langle \boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| \geq \lambda_\infty^\epsilon\right\}\right|,\end{aligned}$$

Thus, with high probability, we have

$$\begin{aligned}|S_1^{t+1}| &\leq \sum_{i\in(S^*)^c}\mathrm{I}\left\{|\boldsymbol{\Xi}_i + \langle \boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| \geq \lambda_\infty^\epsilon\right\} \\ &\leq \sum_{i=1}^{p}\mathrm{I}\left\{|\boldsymbol{\Xi}_i| \geq \lambda_\infty^{\epsilon/2}\right\} + \sum_{i\in(S^*)^c}\mathrm{I}\left\{|\langle \boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^* - \boldsymbol{\beta}^t\rangle| \geq \frac{\epsilon}{2}\lambda_\infty^0\right\} \\ &\leq \frac{s^*}{\log(p/s^*)} + \frac{4\delta_{3s}^2\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2^2}{\epsilon^2\left(\lambda_\infty^0\right)^2},\end{aligned} \tag{58}$$

where the third inequality follows from (27) of Lemma 5.

Combining the above two cases, we complete the proof of Lemma 7.

D. Proof of Corollary 2

Assume that $\tau \in [0,\ s^*/\log(p/s^*)]$, and we can write $s = s^* + \tau \in [s^*,\ s^* + s^*/\log(p/s^*)]$.

Case 1:$\lambda_{s,t+1} \leq \lambda_\infty^\epsilon$.

From (57), for $t \geq \log\left(C_{1,\epsilon}^2 \log\frac{p}{s^*}\right)$, with high probability, we have

$$\begin{aligned}|S_2^{t+1}| &\leq s^*/\log(p/s^*) + \frac{4\delta_{3s}^2\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2^2}{\epsilon^2\left(\lambda_\infty^\epsilon\right)^2} \\ &= O\left(s^*/\log(p/s^*)\right),\end{aligned} \tag{59}$$

where the last inequality follows for the conclusion of Theorem 3. Thus we obtain $|S_1^{t+1}|\vee|S_2^{t+1}| = O\left(s^*/\log(p/s^*)\right) + \tau = O\left(s^*/\log(p/s^*)\right)$.

Case 2: $\lambda_{s,t+1} > \lambda_\infty^\epsilon$.

From (58), for $t \geq \log\left(C_{1,\epsilon}^2 \log\frac{p}{s^*}\right)$, with high probability, we have

$$\begin{aligned}|S_1^{t+1}| &\leq \frac{s^*}{\log(p/s^*)} + \frac{4\delta_{3s}^2\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2^2}{\epsilon^2\left(\lambda_\infty^0\right)^2} \\ &= O\left(s^*/\log(p/s^*)\right),\end{aligned}$$

where the last inequality follows for the conclusion of Theorem 3. Thus we show that $|S_1^{t+1}| \vee |S_2^{t+1}| = O\left(s^*/\log(p/s^*)\right)$

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holds again. Combining the above two cases, we complete the proof of Corollary 2.

E. Proof of Theorem 4

Here we consider the oracle rate under a stronger beta-min condition. And similarly to Theorem 3, we will separate the proof into two cases. However, we now compare our HTP estimator $\boldsymbol{\beta}^t$ to the Oracle OLS estimator $\tilde{\boldsymbol{\beta}}^*$, which is meaningful since this result shows that the HTP estimator converges to the Oracle estimator as we choose s^* as the model size correctly.

By using the oracle estimator $\tilde{\boldsymbol{\beta}}^*$, we can decompose $\boldsymbol{H}^{t+1}$ as

$$\boldsymbol{H}^{t+1} = \tilde{\boldsymbol{\beta}}^* + \boldsymbol{\Phi}(\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t) + \tilde{\boldsymbol{\Xi}},$$

where $\tilde{\boldsymbol{\Xi}}_{(S^*)^c} = \boldsymbol{X}_{(S^*)^c}^\top \left(\mathrm{I}_n - \boldsymbol{X}_{S^*}(\boldsymbol{X}_{S^*}^\top \boldsymbol{X}_{S^*})^{-1}\boldsymbol{X}_{S^*}^\top\right)\boldsymbol{\xi}/n$ and $\tilde{\boldsymbol{\Xi}}_{S^*} = 0$. Note that

$$\begin{aligned}
&\mathrm{pr}\left\{\|\boldsymbol{\beta}^* - \tilde{\boldsymbol{\beta}}^*\|_\infty > \sigma\left(\frac{2\log p}{n}\right)^{1/2}\right\}\\
=&\,\mathrm{pr}\left\{\bigcup_{i\in S^*}\left\{\left|\left((\boldsymbol{X}_{S^*}^\top\boldsymbol{X}_{S^*})^{-1}\boldsymbol{X}_{S^*}^\top\boldsymbol{\xi}\right)_i\right| > \sigma\left(\frac{2\log p}{n}\right)^{1/2}\right\}\right\}\\
\le&\,2s^*\exp\left(-\frac{n}{2\sigma^2}\left(\sigma\left(\frac{2\log p}{n}\right)^{1/2}\right)^2\right)\\
=&\,\frac{2s^*}{p},
\end{aligned} \tag{60}$$

which leads to $\|\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^*\|_\infty \le \sigma\left(2\log p/n\right)^{1/2}$ with high probability. Recall that $\tilde{\lambda}^\epsilon = (1+\epsilon)\sigma(2\log p/n)^{1/2}$. We bound the maximum $|\tilde{\boldsymbol{\Xi}}_i|$ as follows:

$$\begin{aligned}
&\mathrm{pr}\left\{\max_{i\in[p]}|\tilde{\boldsymbol{\Xi}}_i| > (1+\epsilon/2)\left(\frac{2\log p}{n}\right)^{1/2} = \tilde{\lambda}_\infty^{\epsilon/2}\right\}\\
=&\,\mathrm{pr}\left\{\bigcup_{i\in[p]}\left\{|\tilde{\boldsymbol{\Xi}}_i| > (1+\epsilon/2)\left(\frac{2\log p}{n}\right)^{1/2}\right\}\right\}\\
\le&\,2p\exp\left(-(1+\epsilon/2)\log p\right)\\
=&\,p^{-\epsilon/2},
\end{aligned} \tag{61}$$

which leads to $|\tilde{\boldsymbol{\Xi}}_i| \le \tilde{\lambda}_\infty^{\frac{\epsilon}{2}}$. Similar to the proof of Theorem 3, we provide two useful lemmas as follows.

Lemma 8: Assume all the conditions in Theorem 4 hold. Then, with probability tending to 1, we have

$$\begin{aligned}
\|\tilde{\boldsymbol{\beta}}^{t+1} - \tilde{\boldsymbol{\beta}}^*\|_2 \le &\left(1 + 2\sqrt{2} + \frac{4\sqrt{2}}{\epsilon}\right)\delta_{3s}\|\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\|_2 +\\
&2(1+\epsilon)\sigma\left(\frac{\tau\log p}{n}\right)^{1/2}
\end{aligned} \tag{62}$$

Lemma 9: Assume all the conditions in Theorem 4 hold. Then, with probability tending to 1, the type-I error satisfies that

$$|S_1^{t+1}| \le \frac{4\delta_{3s}^2\|\boldsymbol{\beta}^t - \tilde{\boldsymbol{\beta}}^*\|_2^2}{\epsilon^2\left(\tilde{\lambda}_\infty^0\right)^2} + \tau. \tag{63}$$

Proof 11 (Proof of Theorem 4): Denote matrix $\mathbf{P}_{S^*} = \mathbb{I}_n - \boldsymbol{X}_{S^*}(\boldsymbol{X}_{S^*}^\top\boldsymbol{X}_{S^*})^{-1}\boldsymbol{X}_{S^*}^\top$. Note that

$$\begin{aligned}
\|y - \boldsymbol{X}\boldsymbol{\beta}^{t+1}\|_2 &= \|\boldsymbol{X}\tilde{\boldsymbol{\beta}}^* + \mathbf{P}_{S^*}\boldsymbol{\xi} - \boldsymbol{X}\boldsymbol{\beta}^{t+1}\|_2\\
&= \|\boldsymbol{X}(\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^{t+1}) + \mathbf{P}_{S^*}\boldsymbol{\xi}\|_2
\end{aligned}$$

and

$$\|y - \boldsymbol{X}\tilde{\boldsymbol{\beta}}^{t+1}\|_2 = \|\boldsymbol{X}(\tilde{\boldsymbol{\beta}}^* - \tilde{\boldsymbol{\beta}}^{t+1}) + \mathbf{P}_{S^*}\boldsymbol{\xi}\|_2.$$

Since $\|y - \boldsymbol{X}\boldsymbol{\beta}^{t+1}\|_2 \le \|y - \boldsymbol{X}\tilde{\boldsymbol{\beta}}^{t+1}\|_2$, we have

$$\begin{aligned}
&\|\boldsymbol{X}(\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^{t+1})\|_2^2 - 2|\langle\mathbf{P}_{S^*}\boldsymbol{\xi}, \boldsymbol{X}(\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^{t+1})\rangle|\\
\le&\|\boldsymbol{X}(\tilde{\boldsymbol{\beta}}^* - \tilde{\boldsymbol{\beta}}^{t+1})\|_2^2 + 2|\langle\mathbf{P}_{S^*}\boldsymbol{\xi}, \boldsymbol{X}(\tilde{\boldsymbol{\beta}}^* - \tilde{\boldsymbol{\beta}}^{t+1})\rangle|.
\end{aligned} \tag{64}$$

Note that $\mathbf{P}_{S^*}$ projects vector $\boldsymbol{\Xi}$ onto the span space of columns of $\boldsymbol{X}$ indexed by the set $(S^*)^c$. Similar to the proof of Lemma 3, with probability at least $1 - \exp\{-Cs^*\log(p/s^*)\}$, we have

$$\left\langle\mathbf{P}_{S^*}\boldsymbol{\xi}, \frac{\boldsymbol{X}(\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^{t+1})}{\|\boldsymbol{X}(\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^{t+1})\|_2}\right\rangle^2 \le 4|S_1^{t+1}|\log(p/s^*).$$

By (63), we have

$$\left\langle\mathbf{P}_{S^*}\boldsymbol{\xi}, \frac{\boldsymbol{X}(\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^{t+1})}{\|\boldsymbol{X}(\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^{t+1})\|_2}\right\rangle^2 \tag{65}$$

$$\le 4\left(\frac{4\delta_{3s}^2 n\|\boldsymbol{\beta}^t - \tilde{\boldsymbol{\beta}}^*\|_2^2}{\epsilon^2\sigma^2} + \tau\log p\right). \tag{66}$$

Combining (64) and (65), we have

$$\begin{aligned}
\|\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^{t+1}\|_2 \le &\left(\frac{1+\delta_{3s}}{1-\delta_{3s}}\right)^{1/2}\|\tilde{\boldsymbol{\beta}}^* - \tilde{\boldsymbol{\beta}}^{t+1}\|_2 +\\
&\left\{\frac{4}{(1-\delta_{3s})n}\left(\frac{16\delta_{3s}n}{\sigma^2}\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2^2 + 4\tau\log p\right)\right\}^{1/2}.
\end{aligned} \tag{67}$$

From (62) and (67), by some simple algebras, we have

$$\begin{aligned}
&\|\boldsymbol{\beta}^{t+1} - \tilde{\boldsymbol{\beta}}^*\|_2\\
\le&\frac{\sqrt{1+\delta_{3s}}\left(1 + 2\sqrt{2} + \frac{4\sqrt{2}}{\epsilon}\right)\delta_{3s} + 8\sqrt{\delta_{3s}}}{\sqrt{1-\delta_{3s}}}\|\boldsymbol{\beta}^t - \tilde{\boldsymbol{\beta}}^*\|_2 +\\
&C_1\sigma\left(\frac{\tau\log p}{n}\right)^{1/2}\\
\le&\widetilde{\gamma}_{3s,\epsilon}\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_2 + C_1\sigma\left(\frac{\tau\log p}{n}\right)^{1/2},
\end{aligned}$$

where the second inequality follows from $\widetilde{\gamma}_{3s,\epsilon} < 1$. By using the above inequality repeatedly, we have

$$\begin{aligned}
&\|\boldsymbol{\beta}^{t+1} - \tilde{\boldsymbol{\beta}}^*\|_2\\
\le&\widetilde{\gamma}_{3s,\epsilon}^t\|\boldsymbol{\beta}^0 - \tilde{\boldsymbol{\beta}}^*\|_2 + C_1\sigma\left(\frac{\tau\log p}{n}\right)^{1/2}\\
\le&\widetilde{\gamma}_{3s,\epsilon}^t\sigma\left(\frac{s^*\log p/s^*}{n}\right)^{1/2} + C_1\sigma\left(\frac{\tau\log p}{n}\right)^{1/2}.
\end{aligned}$$

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1) Proof of Lemma 8: Define $\tilde{A}_i^{t+1} := \tilde{\boldsymbol{\Xi}}_i + \langle \boldsymbol{\Phi}_i^\top, \tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\rangle$. Note that $\sqrt{\sum_{i\in S_2^{t+1}} \left(\boldsymbol{H}_i^{t+1}\right)^2 \mathrm{I}\left\{|\boldsymbol{H}_i^{t+1}| < \lambda_{s,t+1}\right\}} < \sqrt{\sum_{i\in S_1^{t+1}} (\tilde{A}_i^t)^2}$. Similar to (51), we have

$$\begin{aligned}
&\|\tilde{\boldsymbol{\beta}}^{t+1} - \tilde{\boldsymbol{\beta}}^*\|_2 \\
&= \Bigg\{ \sum_{i\in S_1^{t+1}\cup S_3^{t+1}} (\tilde{A}_i^{t+1})^2 + \sum_{i\in S_2^{t+1}} \\
&\left\{\langle \boldsymbol{\Phi}_i^\top, \boldsymbol{\beta}^t - \tilde{\boldsymbol{\beta}}^*\rangle^2 + \left(\boldsymbol{H}_i^{t+1}\right)^2 \mathrm{I}\{|\boldsymbol{H}_i^{t+1}| < \lambda_{s,t+1}\}\right\}\Bigg\}^{1/2} \\
&\le \|\langle \boldsymbol{\Phi}_{S_2^{t+1}\cup S_3^{t+1}}^\top, \boldsymbol{\beta}^t - \tilde{\boldsymbol{\beta}}^*\rangle\|_2 + \sqrt{2\sum_{i\in S_1^{t+1}} (A_i^{t+1})^2},
\end{aligned} \tag{68}$$

where the first inequality follows from the definition of sets S_1, S_2, S_3 and the triangle inequality.

Case 1: $\lambda_{s,t+1} \ge \tilde{\lambda}_\infty^\epsilon$. With probability approaching to 1, we have

$$\begin{aligned}
&\sqrt{\sum_{i\in S_1^{t+1}} (\tilde{A}_i^{t+1})^2} \\
&\le \left(\sum_{i\in S_1^{t+1}} \tilde{\boldsymbol{\Xi}}_i^2 \mathrm{I}\left\{|\tilde{\boldsymbol{\Xi}}_i + \langle \boldsymbol{\Phi}_i^\top, \tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\rangle| \ge \tilde{\lambda}_\infty^\epsilon\right\}\right)^{1/2} + \\
&\|\langle \boldsymbol{\Phi}_{S_1^{t+1}}^\top, \tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\rangle\|_2 \\
&= \left(\sum_{i\in S_1^{t+1}} \tilde{\boldsymbol{\Xi}}_i^2 \mathrm{I}\left\{|\tilde{\boldsymbol{\Xi}}_i| \ge \tilde{\lambda}_\infty^{\frac{\epsilon}{2}}\right\}\right)^{1/2} + \|\langle \boldsymbol{\Phi}_{S_1^{t+1}}^\top, \tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\rangle\|_2 + \\
&\left(\sum_{i\in S_1^{t+1}} \tilde{\boldsymbol{\Xi}}_i^2 \mathrm{I}\left\{|\tilde{\boldsymbol{\Xi}}_i| \le \tilde{\lambda}_\infty^{\frac{\epsilon}{2}}, |\langle \boldsymbol{\Phi}_i^\top, \tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\rangle| \ge \frac{\epsilon\tilde{\lambda}_\infty^0}{2}\right\}\right)^{1/2} \\
&\le 2\left(1 + \frac{1}{\epsilon}\right)\delta_{3s}\|\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\|_2,
\end{aligned}$$

where the last inequality follows from (61). Therefore, we have

$$\begin{aligned}
\|\tilde{\boldsymbol{\beta}}^{t+1} - \tilde{\boldsymbol{\beta}}^*\|_2 &\le \|\langle \boldsymbol{\Phi}_{S_2^{t+1}\cup S_3^{t+1}}^\top, \tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\rangle\|_2 + \\
&\sqrt{2\sum_{i\in S_1^{t+1}} \left(\tilde{A}_i^t\right)^2} \\
&\le \left(1 + 2\sqrt{2} + \frac{2\sqrt{2}}{\epsilon}\right)\delta_{3s}\|\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\|_2.
\end{aligned}$$

Case 2: $\lambda_{s,t+1} < \tilde{\lambda}_\infty^\epsilon$. Consider $S_{11}^{t+1} := \{i \in S_1^{t+1} : |\tilde{A}_i^{t+1}| \le \tilde{\lambda}_\infty^\epsilon\}$. For any $j \in S_2^{t+1}$, we have

$$\begin{aligned}
&|\tilde{\boldsymbol{\beta}}_j^*| - |\langle \boldsymbol{\Phi}_j^\top, \tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\rangle| \\
&\le |\tilde{\boldsymbol{\beta}}_j^* + \langle \boldsymbol{\Phi}_j^\top, \tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\rangle| = |\boldsymbol{H}_j^{t+1}| < \lambda_{s,t+1} < \tilde{\lambda}_\infty^\epsilon.
\end{aligned}$$

Note that $\tilde{\boldsymbol{\Xi}}_i = 0, \ \forall j \in S_2^{t+1}$, thus, by (60) we have

$$\begin{aligned}
|\tilde{A}_j^{t+1}| = |\langle \boldsymbol{\Phi}_j^\top, \tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\rangle| &> |\tilde{\boldsymbol{\beta}}_j^*| - \tilde{\lambda}_\infty^\epsilon \\
&\ge |\boldsymbol{\beta}_j^*| - \sigma\left(\frac{2\log p}{n}\right)^{1/2} - \tilde{\lambda}_\infty^\epsilon \\
&\ge \epsilon\tilde{\lambda}_\infty^0 \ \ge \ \frac{\epsilon}{2}\tilde{\lambda}_\infty^\epsilon.
\end{aligned}$$

Since $|S_{11}^{t+1}| \le |S_1^{t+1}| = |S_2^{t+1}| + \tau, \ \tau \in [0, \ 3s^*]$, we have

$$\begin{aligned}
\|\tilde{A}_{S_{11}^{t+1}}^{t+1}\|_2 &\le \sqrt{\sum_{j\in S_2^{t+1}} \left(\tilde{\lambda}_\infty^\epsilon\right)^2} + \sqrt{\tau\left(\tilde{\lambda}_\infty^\epsilon\right)^2} \\
&\le \frac{2}{\epsilon}\|\tilde{A}_{S_2^{t+1}}^{t+1}\|_2 + (1+\epsilon)\sigma\left(\frac{2\tau\log p}{n}\right)^{1/2} \\
&\le \frac{2}{\epsilon}\delta_{3s}\|\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\|_2 + (1+\epsilon)\sigma\left(\frac{2\tau\log p}{n}\right)^{1/2}.
\end{aligned}$$

As a consequence, we have

$$\begin{aligned}
\|\tilde{A}_{S_1^{t+1}}^{t+1}\|_2 &\le \|\tilde{A}_{S_{11}^{t+1}}^{t+1}\|_2 + \|\tilde{A}_{S_1^{t+1}\setminus S_{11}^{t+1}}^{t+1}\|_2 \\
&\le \left(2 + \frac{4}{\epsilon}\right)\delta_{3s}\|\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\|_2 + \\
&(1+\epsilon)\sigma\left(\frac{2\tau\log p}{n}\right)^{1/2}.
\end{aligned}$$

Therefore, we have

$$\begin{aligned}
\|\tilde{\boldsymbol{\beta}}^{t+1} - \tilde{\boldsymbol{\beta}}^*\|_2 &\le \|\langle \boldsymbol{\Phi}_{S_2^{t+1}\cup S_3^{t+1}}^\top, \tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\rangle\|_2 + \\
&\sqrt{2\sum_{i\in S_1^{t+1}} \left(\tilde{A}_i^t\right)^2} \\
&\le \left(1 + 2\sqrt{2} + \frac{4\sqrt{2}}{\epsilon}\right)\delta_{3s}\|\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\|_2 + \\
&2(1+\epsilon)\sigma\left(\frac{\tau\log p}{n}\right)^{1/2}.
\end{aligned}$$

According to **case 1** and **case 2**, with high probability, we finally get that

$$\begin{aligned}
\|\tilde{\boldsymbol{\beta}}^{t+1} - \tilde{\boldsymbol{\beta}}^*\|_2 &\le \left(1 + 2\sqrt{2} + \frac{4\sqrt{2}}{\epsilon}\right)\delta_{3s}\|\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^t\|_2 + \\
&2(1+\epsilon)\sigma\left(\frac{\tau\log p}{n}\right)^{1/2}, \ \forall t \ge 0.
\end{aligned}$$

2) Proof of Lemma 9: Proof 12: The proof of Lemma 9 is similar to Lemma 7. Recall that $\tilde{\lambda}^\epsilon = (1+\epsilon)\sigma(2\log p/n)^{1/2}$, we only change (57) and (58) to

$$\sum_{i\in S^*} \mathrm{I}\left\{|\boldsymbol{\Xi}_i| \ge \frac{\epsilon\tilde{\lambda}^\epsilon}{2}\right\} = o(1) \tag{69}$$

and

$$\sum_{i\in [p]} \mathrm{I}\left\{|\boldsymbol{\Xi}_i| \ge \frac{\epsilon\tilde{\lambda}^\epsilon}{2}\right\} = o(1) \tag{70}$$

hold with probability larger than $1 - p^{-\delta}$ for some constant δ independent to n, p, s.

F. Proof of Corollary 3

Proof 13: Given $s = s^*$ and $t \ge 2\log\left((C_2/\epsilon)^2 s^*\right)$, from the conclusion of Theorem 4, we have $\|\boldsymbol{\beta}^t - \tilde{\boldsymbol{\beta}}^*\|_2 < \epsilon\sigma(2\log p/n)^{1/2}$, which leads to

$$\begin{aligned}
\|\boldsymbol{\beta}^t - \boldsymbol{\beta}^*\|_\infty &\le \|\boldsymbol{\beta}^t - \tilde{\boldsymbol{\beta}}^*\|_\infty + \|\tilde{\boldsymbol{\beta}}^* - \boldsymbol{\beta}^*\|_\infty \\
&< (1+\epsilon)\sigma\left(\frac{2\log p}{n}\right)^{1/2}.
\end{aligned} \tag{71}$$

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Therefore, for $\forall i \in S^*$, we have $|\beta_i^t| > |\beta_i^*| - (1+\epsilon)\sigma(2n^{-1}\log p)^{1/2} \geq (1+\epsilon)\sigma(2n^{-1}\log p)^{1/2}$. Besides, from (71) we can also bound all $|\beta_j^t|$, $\forall j \in (S^*)^c$, that is,

$$|\beta_j^t| = |\beta_j^t - \beta_j^*| \leq \|\beta^t - \beta^*\|_\infty < (1+\epsilon)\sigma\left(\frac{2\log p}{n}\right)^{1/2}.$$

Thus we show that all signal estimations of the true support set S^* are uniformly larger than those (potential) of $(S^*)^c$. Since we only select the top s^* largest absolute signals in each iteration, it implies that $S^t = S^*$ holds for all $t \geq \log\left((C_2/\epsilon)^2 s^*\right)$, with probability approaching to 1.

G. Proof of Theorem 5

First, we prove that a correct model size $s = s^*$ leads to a large minimum signal (see (11)), while a large model size $s > s^*$ leads to a small one, i.e., (12), which provides a theoretical support for achieving exact recovery.

Proof 12:

Proof of (11):

According to Theorem 4, for $t \geq 2\log\left(C_2^2 s^*\right)$, we have

$$\begin{aligned}\sigma\left(\frac{2\log p}{n}\right)^{1/2} &\geq \|\beta^t - \tilde{\beta}^*\|_2 \geq \|\beta^t - \tilde{\beta}^*\|_\infty \\ &\geq |\tilde{\beta}_i^*| - |\beta_i^t|,\ \forall i \in S^*. \end{aligned} \tag{72}$$

From (60), we have $|\tilde{\beta}_i^*| \geq |\beta_i^*| - \sigma\left(2\log p/n\right)^{1/2}$. Thus, combining with (72), for $\forall i \in S^*$, we have

$$\begin{aligned}&\sigma\left(\frac{2\log p}{n}\right)^{1/2} \\ &\geq |\tilde{\beta}_i^*| - |\beta_i^t| \geq |\beta_i^*| - \sigma\left(\frac{2\log p}{n}\right)^{1/2} - |\beta_i^t|,\end{aligned}$$

which implies that $\lambda_{s,t} \geq ((C_1 - 2) \vee 2)\,\sigma(2\log p/n)^{1/2}$.

Proof of (12):

We prove (12) by contradiction. For $\forall t \geq 2\log\left(C_2^2 s^*/\tau\right)$ and $\tau \in [3s^*]$, from Theorem 4, we have

$$\|\beta^t - \tilde{\beta}^*\|_2 \leq (7+6\epsilon)\,\sigma\left(\frac{2\tau\log p}{n}\right)^{1/2}. \tag{73}$$

Now we assume that there exists $t_0 \geq 2\log\left(C_2^2 s^*/\tau\right)$ such that $\lambda_{t_0+1} \geq 2\sigma(2\log p/n)^{1/2}$. Thus, with high probability, we have

$$\begin{aligned}&|S_1^{t_0+1}| \\ &= \sum_{i\in(S^*)^c} \mathrm{I}\left\{|\tilde{\Xi}_i + \langle \Phi_i^\top, \tilde{\beta}^* - \beta^{t_0}\rangle| \geq \lambda_{t_0+1}\right\} \\ &\leq \sum_{i\in(S^*)^c} \mathrm{I}\left\{|\tilde{\Xi}_i + \langle \Phi_i^\top, \tilde{\beta}^* - \beta^{t_0}\rangle| \geq 2\sigma\left(\frac{2\log p}{n}\right)^{1/2}\right\} \\ &\leq \sum_{i=1}^{p} \mathrm{I}\left\{|\tilde{\Xi}_i| \geq \frac{3\sigma}{2}\left(\frac{2\log p}{n}\right)^{1/2}\right\} + \\ &\quad \sum_{i\in(S^*)^c} \mathrm{I}\left\{|\langle \Phi_i^\top, \tilde{\beta}^* - \beta^{t_0}\rangle| \geq \frac{\sigma}{2}\left(\frac{2\log p}{n}\right)^{1/2}\right\} \\ &\leq \frac{4n\delta_{3s}^2\|\beta^{t_0} - \tilde{\beta}^*\|_2^2}{2\sigma^2\log p}. \end{aligned} \tag{74}$$

where last inequality follows from (61). Note that $\epsilon \in (0,1)$ and $\delta_{3s} < \epsilon^2 \wedge \frac{7-4\sqrt{3}}{12}$, which implies that

$$\begin{aligned}|S_1^{t_0+1}| &\leq \frac{4\, n\delta_{3s}^2\|\beta^{t_0} - \tilde{\beta}^*\|_2^2}{2\sigma^2\log p} \leq 4\delta_{3s}^2(7+6\epsilon)^2\tau < 676\delta_{3s}^2\tau \\ &< \tau.\end{aligned}$$

Since $s = s^* + \tau$, we have $|S_1^{t_0+1}| \geq \tau$, which shows that (74) leads to an absurd. Thus, we prove (12) holds.

H. Proof of Corollary 4

Proof 15: We separate the proof into two cases based on whether $\beta^* \in \Omega^p_{s^*,\lambda}$, where $\lambda \geq \left(C_1 \vee 4\right)\sigma\left(2\log p/n\right)^{1/2}$ and constant C_1 is determined in (6).

Case 1: $\beta^* \in \Omega^p_{s^*,\lambda}$. Then, based on Theorem 5, we achieve exact recovery and $\hat{s} = s^*$. Therefore, by using Theorem 4 and choosing $t > 2\log\left(C_2^2\log p\right)$, we get $\|\hat{\beta}^{(\hat{s})} - \tilde{\beta}^*\|_2 \leq \sigma(s^*/n)^{1/2}$. Besides, note that $\tilde{\beta}^*$ is the OLS estimator based on the true support set S^*, we obtain

$$\begin{aligned}\|\tilde{\beta}^* - \beta^*\|_2^2 &= \|(X_{S^*}^\top X_{S^*})^{-1}X_{S^*}^\top \xi\|_2^2 \\ &\leq \|(X_{S^*}^\top X_{S^*})^{-1}\|_2^2\|X_{S^*}^\top \xi\|_2^2 \\ &\overset{(i)}{\leq} \left(\frac{1}{(1-\delta_s)n}\right)^2 4s^* n\sigma^2 \\ &\overset{(ii)}{\leq} 16\sigma^2\frac{s^*}{n}\end{aligned} \tag{75}$$

holds with probability higher than $1 - \exp(-s^*/4)$, where inequality (i) uses RIP and (53), inequality (ii) uses $\delta_s \leq 1/2$. By triangle inequality, we have $\|\hat{\beta}^{(\hat{s})} - \beta^*\|_2 \leq \|\hat{\beta}^{(\hat{s})} - \tilde{\beta}^*\|_2 + \|\tilde{\beta}^* - \beta^*\|_2 \leq 5\sigma\,(s^*/n)^{1/2}$, which yields that

$$\begin{aligned}\|\hat{\beta}^{(\bar{s})} - \hat{\beta}^{(\hat{s})}\|_2 &\leq \|\hat{\beta}^{(\bar{s})} - \beta^*\|_2 + \|\hat{\beta}^{(\hat{s})} - \beta^*\|_2 \\ &\overset{(i)}{\leq} 5\sigma\left(\frac{s^*}{n}\right)^{1/2} + C_1\sigma\left\{\frac{s^*\log(p/s^*)}{n}\right\}^{1/2} \\ &\leq (5 + C_1)\sigma\left\{\frac{s^*\log(p/s^*)}{n}\right\}^{1/2} \\ &\overset{(ii)}{\leq} \sqrt{2}(5+C_1)\sigma\left\{\frac{\hat{s}\log(p/\hat{s})}{n}\right\}^{1/2},\end{aligned} \tag{76}$$

where inequality (i) follows the error bound from (6), and inequality (ii) follows $\hat{s} \geq s^*/2$ in (7). Based on (76), we will choose $\hat{\beta} = \hat{\beta}^{(\hat{s})}$ with probability tending to 1, which leads to $\|\hat{\beta} - \beta^*\|_2 \leq 5\sigma\left(\frac{s^*}{n}\right)^{1/2}$.

Case 2: $\beta^* \notin \Omega^p_{s^*,\lambda}$. We again separate this case into two subcases. If $\|\hat{\beta}^{(\bar{s})} - \hat{\beta}^{(\hat{s})}\|_2^2 \leq C\sigma^2\frac{\hat{s}\log(p/\hat{s})}{n}$, we choose $\hat{\beta} = \hat{\beta}^{(\hat{s})}$ and thus

$$\begin{aligned}&\|\hat{\beta} - \beta^*\|_2 \\ \leq&\|\hat{\beta}^{(\bar{s})} - \hat{\beta}^{(\hat{s})}\|_2 + \|\hat{\beta}^{(\bar{s})} - \beta^*\|_2 \\ \overset{(i)}{\leq}& \sqrt{C}\sigma\left\{\frac{\hat{s}\log(p/\hat{s})}{n}\right\}^{1/2} + C_1\sigma\left\{\frac{s^*\log(p/s^*)}{n}\right\}^{1/2} \\ \overset{(ii)}{\leq}& \left(\sqrt{2C} + C_1\right)\sigma\left\{\frac{s^*\log(p/s^*)}{n}\right\}^{1/2},\end{aligned}$$

where inequality (i) and (ii) use the results (6) and (5) respectively.

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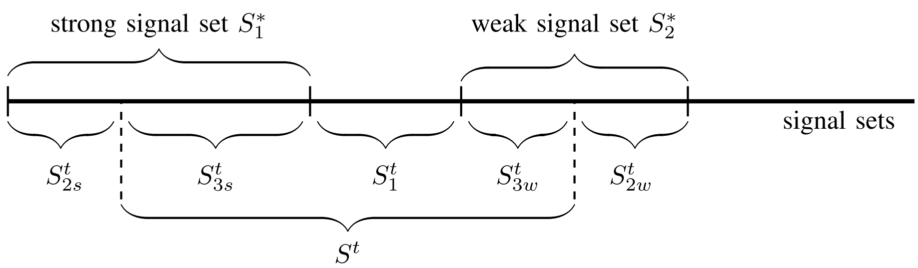

Fig. 9. The illustration of these signal sets, where S_1^*, S_2^* are true strong signal set and weak signal set respectively. In the t-th iteration, $S^t = S_1^t \cup S_{3s}^t \cup S_{3w}^t$ is the estimation set, while S_{2s}^t and S_{2w}^t are the type-II error, i.e., the undiscovered true signal sets respectively. Besides, we note that $|S_1^*| = s_1^*$, $|S_2^*| = s_2^*$ and $|S^t| = s_1^* + \tau$.

On the other hand, if $\|\hat{\boldsymbol{\beta}}^{(\tilde{s})} - \hat{\boldsymbol{\beta}}^{(\hat{s})}\|_2^2 > C\sigma^2 \frac{\hat{s}\log(p/\hat{s})}{n}$, we choose $\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^{(\hat{s})}$. Thus $\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2 \le C_1\sigma\left\{\frac{s^*\log(p/s^*)}{n}\right\}^{1/2}$ by (6).

Finally, based on the two cases, we complete the proof of Corollary 4.

I. Proof of Theorem 6

The proof of Theorem 6 is similar to the proof of Theorem 3. We just need to divide the analysis on support set S^* into S_1^* and S_2^*, respectively. The detailed separation of S^* and S^t are shown in Figure 9. Here we use similar notations in the proof of Theorem 4. We separate the proof into two cases.

For the set $S_{2s}^{t+1} \cup S_{3s}^{t+1} \cup S_{2w}^{t+1}$: Note that $S_1^* \cup S_{2w}^{t+1} = S_{2s}^{t+1} \cup S_{3s}^{t+1} \cup S_{2w}^{t+1}$, with high probability, we have

$$\begin{aligned}
&\|(\boldsymbol{\beta}^* - \boldsymbol{\beta}^{t+1})_{S_1^*\cup S_{2w}^{t+1}}\|_2 \\
=& \left\{\sum_{i\in S_s^*}\left(-\boldsymbol{H}_i^{t+1}\mathbf{1}\left(|\boldsymbol{H}_i^{t+1}| < \lambda_{s,t+1}\right) + \boldsymbol{A}_i^{t+1}\right)^2 + \sum_{j\in S_{2w}^{t+1}} (\boldsymbol{\beta}_j^*)^2\right\}^{1/2} \\
\le& \sqrt{\sum_{i\in S_1^*}\left(\boldsymbol{A}_i^{t+1}\right)^2} + \sqrt{\sum_{i\in S_{2s}^{t+1}}\left(\boldsymbol{H}_i^{t+1}\right)^2 + \sum_{j\in S_{2w}^{t+1}}(\boldsymbol{\beta}_j^*)^2} \\
\le& \delta_{3s^*}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2 + 2\sqrt{\frac{s_1^*}{n}} + \sqrt{2}\|\boldsymbol{\beta}^*_{S_2^*}\|_2 + \\
&\|(\boldsymbol{\beta}^* - \boldsymbol{\beta}^{t+1})_{S_1^{t+1}\cup S_{3w}^{t+1}}\|_2. \qquad (77)
\end{aligned}$$

For the set $S_1^{t+1} \cup S_{3w}^{t+1}$: Since $S_1^{t+1} \cup S_{3w}^{t+1} \subseteq S^{t+1}$, we derive that $|H_i^{t+1}| \ge \lambda_{s,t+1}$ holds for $\forall i \in S_1^{t+1} \cup S_{3w}^{t+1}$, which leads that

$$\begin{aligned}
&\|(\boldsymbol{\beta}^* - \boldsymbol{\beta}^{t+1})_{S_1^{t+1}\cup S_{3w}^{t+1}}\|_2 \\
\le& \delta_{3s^*}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2 + \sqrt{\sum_{i\in S_1^{t+1}\cup S_{3w}^{t+1}} \boldsymbol{\Xi}_i^2\mathbf{1}(|\boldsymbol{H}_i^{t+1}| \ge \lambda_{s,t+1})}. \qquad (78)
\end{aligned}$$

Similar to the proof of Theorem 4, we divide the analysis of (78) in two cases: $\lambda_{s,t+1} \ge (1+\epsilon)\sigma(\log p/n)^{1/2}$ and $\lambda_{s,t+1} < (1+\epsilon)\sigma(\log p/n)^{1/2}$. (Here the constant C_1 defined in S_1^* is larger than $1+\epsilon$. We use $1+\epsilon$ just for the simplicity of proof.) By a similar analysis, with high probability, we obtain

$$\begin{aligned}
&\|(\boldsymbol{\beta}^* - \boldsymbol{\beta}^{t+1})_{S_1^{t+1}\cup S_{3w}^{t+1}}\|_2 \\
\le& \left(2\sqrt{2} + \frac{2\sqrt{2}}{\epsilon}\right)\delta_{3s}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2 + \|\boldsymbol{\beta}^*{}_{S_2^*}\|_2 + \\
&(2\sqrt{2} + \sqrt{2}\epsilon)\sqrt{\frac{\tau\log(p/s^*)}{n}} + \left(3 + \frac{4}{\epsilon}\right)\sqrt{\frac{s_1^*}{n}}. \qquad (79)
\end{aligned}$$

The total ℓ_2 error: Combining (78) and (79) together, with high probability, we derive that

$$\begin{aligned}
\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^{t+1}\|_2 \le& \|(\boldsymbol{\beta}^* - \boldsymbol{\beta}^{t+1})_{S_s^*\cup S_{2w}^{t+1}}\|_2 + \\
&\|(\boldsymbol{\beta}^* - \boldsymbol{\beta}^{t+1})_{S_1^{t+1}\cup S_{3w}^{t+1}}\|_2 \\
\le& \left(1 + 4\sqrt{2} + \frac{4\sqrt{2}}{\epsilon}\right)\delta_{3s^*}\|\boldsymbol{\beta}^* - \boldsymbol{\beta}^t\|_2 + \\
&(4\sqrt{2} + 2\sqrt{2}\epsilon)\sqrt{\frac{\tau\log(p/s^*)}{n}} + \\
&3\|\boldsymbol{\beta}^*{}_{S_2^*}\|_2 + \left(8 + \frac{8}{\epsilon}\right)\sqrt{\frac{s_1^*}{n}}.
\end{aligned}$$

For a small δ_{3s^*} and a sufficiently large t, we prove Theorem 6.

J. Proof of Corollary 6

Here we separate the proof into two cases. Firstly, when $\|\boldsymbol{\beta}^*_{S_2^*}\|_2 \le C_1\sigma(\log p/n)^{1/2}$, Theorem 5 can be obtained directly for $s = |S_1^*|$. Therefore, we can exactly identify $|S_1^*|$ with high probability by FAHTP. On the other hand, for $\|\boldsymbol{\beta}^*_{S_2^*}\|_2 \ge C_1\sigma(\log p/n)^{1/2}$, Theorem 2 can be applied. The error bound $C_2\left(\sigma^2\frac{|S_1^*|\log p}{n} + \|\boldsymbol{\beta}^*_{S_2^*}\|_2^2\right)$ is established for this case.

Additional Simulation Studies

K. The Selection of Hyperprameter K

In this section, we employ a simulation-based approach to determine the optimal hyperparameter K for the criterion defined in (4). To ensure consistency and comparability, we replicate the experimental setup outlined in Section V-C, utilizing a sample size of $n = 300$, a sparsity of $s^* = 20$ and a noise standard deviation of $\sigma = 5$. The dimensionality p is systematically varied across three levels: 1000, 1500, and 2000. For each configuration, the hyperparameter K is systematically evaluated by incrementing it from 1 to 10 in unit steps. The empirical performance of the criterion is then assessed across these K values to identify the setting that yields the most robust and reliable results.

As shown in Figure 10, when K is small, FAHTP tends to select many redundant variables, leading to poor estimation errors and suboptimal variable selection performance. When K is in the range of 3–5, FAHTP achieves stable and efficient

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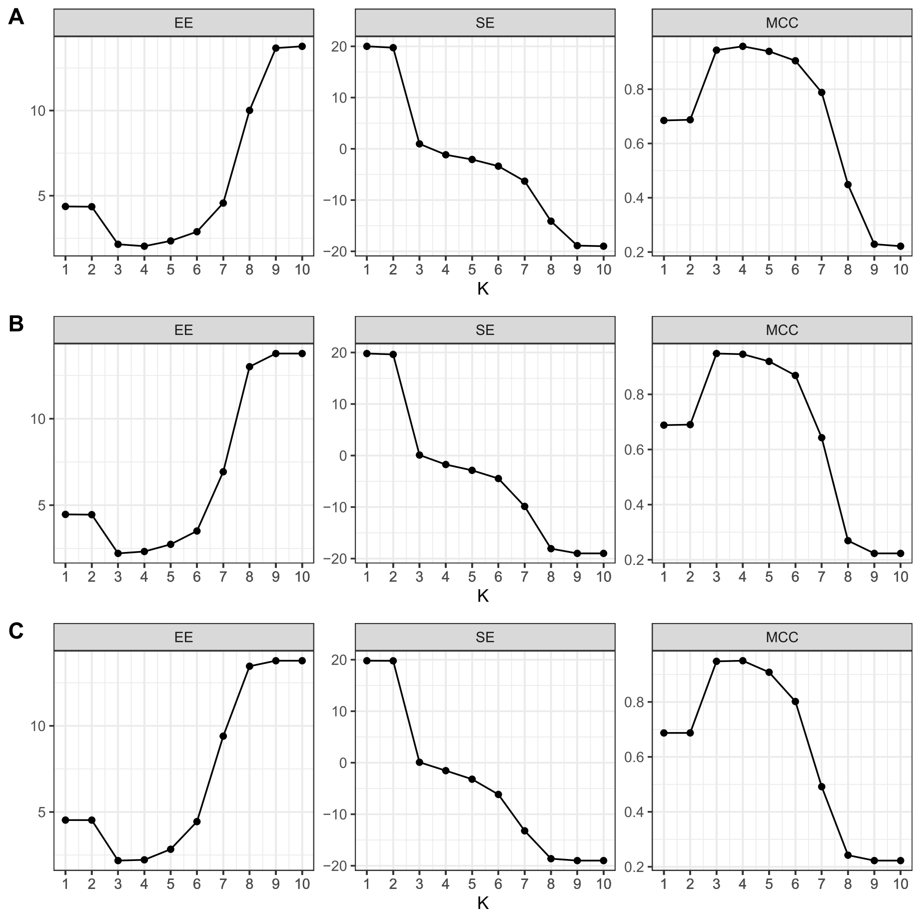

Fig. 10. The empirical performance of FAHTP at increasing K Values. Figure (A) $p = 1,000$. Figure (B) $p = 1,500$. Figure (C) $p = 2,000$.

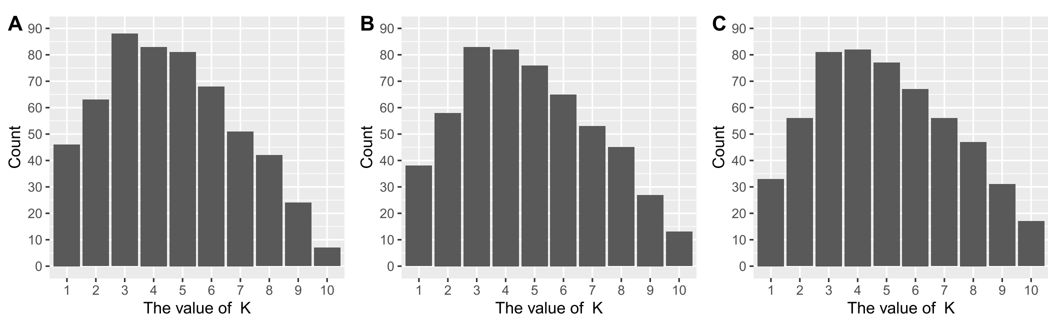

Fig. 11. The barplot of the optimal values of K selected by 5-fold cross-validation. Figure (A) $p = 1,000$. Figure (B) $p = 1,500$. Figure (c) $p = 2,000$.

performance. Notably, in subplots (B) and (C), the estimator with $K = 3$ yields the best results. However, as K further increases, FAHTP underfits, omitting critical variables. Based on numerical experiments, FAHTP achieves stable performance for $K \in [3,5]$, and we recommend setting $K = 3$ for its efficiency and robustness.

To determine the optimal K value empirically, we utilize 5-fold cross-validation in comprehensive simulations ($n = 300$, $\sigma = 5$, $p \in \{1000, 1500, 2000\}$), with 100 replications per condition. Figure 11 presents the resulting distribution of selected K values. Figure 11 reveals that values of K within the [3, 5] range are most frequently selected during cross-validation, indicating that this interval provides optimal empirical performance for FAHTP. This observation shows strong consistency with our preceding simulation results.

To further validate these findings, we performed comparative analyses contrasting fixed K (=3) and cross-validated K selection approaches. Under fixed parameters ($p = 2000, \sigma = 5$), we systematically evaluated performance across varying sample sizes ($n \in [200, 1000]$) while holding other parameters constant.

While Figure 12 reveals comparable empirical performance between the two strategies, the CV-tuned method gradually outperforms in variable selection with increasing sample size. Due to its significant computational cost, we advocate for the fixed K approach in most applications.

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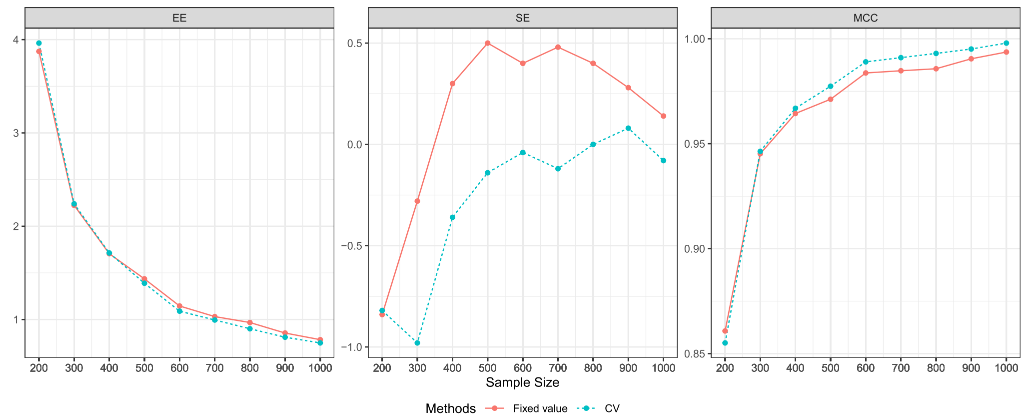

Fig. 12. Empirical evaluation of FAHTP: Fixed $K = 3$ versus 5-fold CV-tuned K selection.

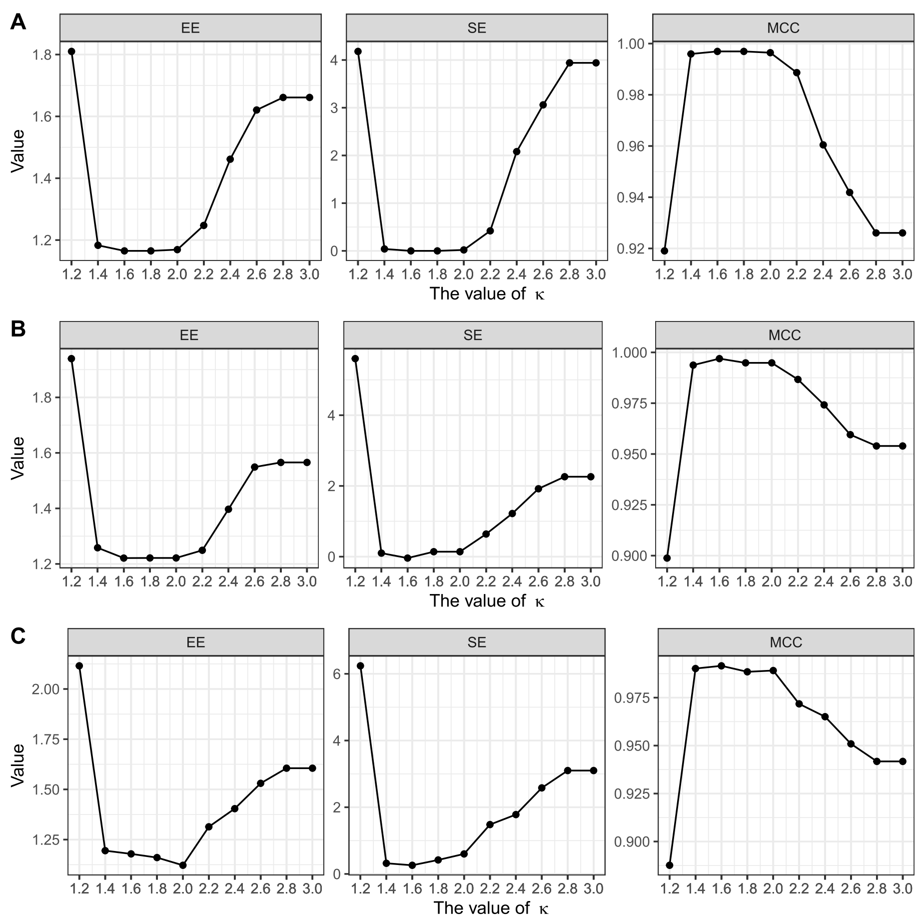

Fig. 13. The empirical performance of FAHTP at increasing κ Values. Figure (A) $p = 1,000$. Figure (B) $p = 1,500$. Figure (C) $p = 2,000$.

L. The Selection of Hyperprameter κ

This section optimizes the hyperparameter κ in Algorithm 2 via simulation. We set $n = 200$ and $\sigma = 3$, and the remaining settings are the same as Section K, κ is tested at increments of 0.2 from 1.2 to 3.0 to identify the robust configuration that achieves the best performance. From Figure 13, we find that when $\kappa \in [1.4, 2.0]$, FAHTP achieves efficient and robust empirical performances. On the other hand, performance degrades progressively as κ deviates from this range, with both higher estimation error and lower values of MCC yielding suboptimal results. Based on these systematic evaluations, we recommend $\kappa = 2$ as the preferred parameter choice, as it consistently provides the most reliable performance across all tested scenarios.

For applications demanding further precision, cross-validation can be employed to fine-tune the value of κ. We implement 5-fold cross-validation (CV) on extended simulations ($n = 200$, $\sigma = 5$, $p \in \{1000, 1500, 2000\}$), retaining 100

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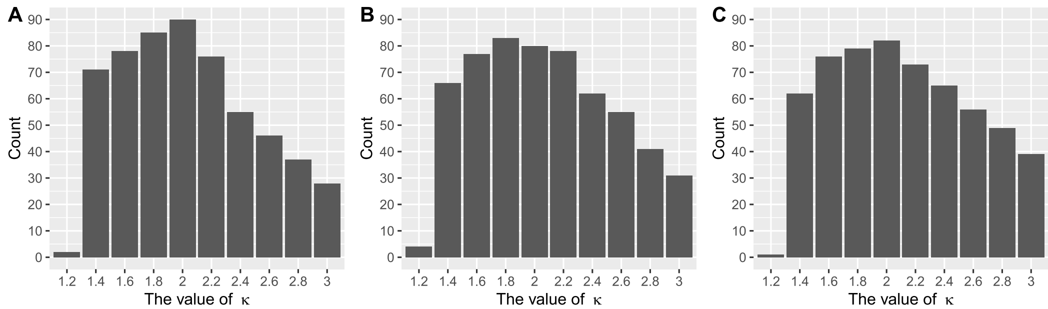

Fig. 14. The barplot of the optimal values of κ selected by 5-fold cross-validation. Figure (A) $p = 1,000$. Figure (B) $p = 1,500$. Figure (c) $p = 2,000$.

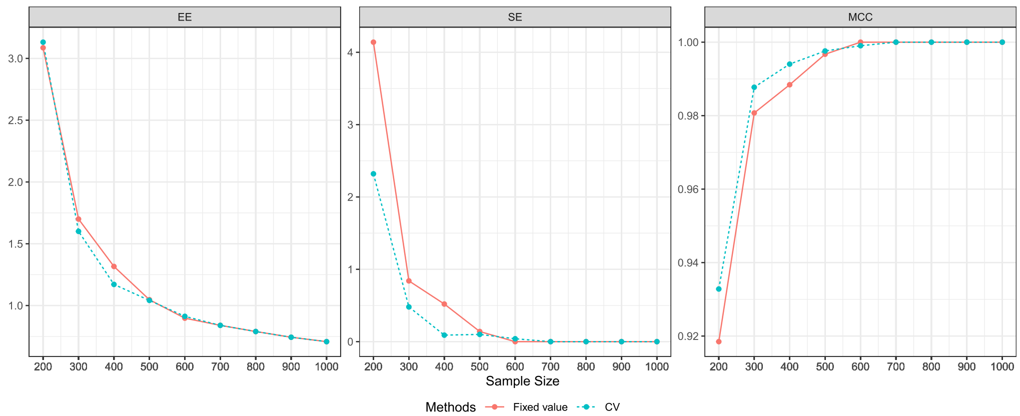

Fig. 15. Empirical evaluation of FAHTP: Fixed $\kappa = 2$ versus 5-fold CV-tuned κ selection.

replications per condition. Figure 14 shows the CV-selected κ distribution:

Figure 14 demonstrates that κ values within [1.6, 2] are selected most frequently during cross-validation, suggesting this range yields optimal empirical performance for FAHTP. This finding aligns consistently with our earlier simulation results.

In addition, to further validate this observation, we conducted comparative analyses between fixed κ (=2) and cross-validated κ selection strategies. Using $p = 2000$ and $\sigma = 5$, we systematically vary sample sizes ($n \in [200, 1000]$) while maintaining other settings unchanged.

From Figure 15, the empirical performance of the two strategies was found to be remarkably similar. Specifically, cross-validation demonstrated superior performance when the sample size was relatively small; however, as the sample size increased, the performance of both strategies converged. Given that cross-validation incurs a notable computational overhead compared to using a fixed κ value, we recommend employing the fixed threshold approach in practical applications. Nevertheless, for scenarios requiring more refined results, cross-validation can be utilized as an adjustment method.

M. The Comparison With Other Best Subset Selection Methods

In this section, we compare our results with other best subset selection methods. We summarize the comparative methods as follows.

TABLE III
THE DETAIL OF THE COMPARATIVE METHODS

Method	Tuning method	Package
ABESS-IC	criterion (4)	`abess`[52]
ABESS-SIC	SIC[6]	`abess`[52]
CD-IC	criterion (4)	`L0Learn` [53]
CDPSI-IC	criterion (4)	`L0Learn` [53]

We set $p = 3000$, $s = 30$. The remaining setting is similar to Section V-C. The computational results with 100 repetitions are shown in Table III.

From Table IV, we see that FAHTP achieves the best performance on estimation error and MCC, which shows its superiority to the state-of-the-art methods for best subset selection. Moreover, FAHTP can control the FPR at a relatively low level, which shows the good performance of our methods on support recovery.

N. The Comparison of IHT, HTP and Lasso

We implement FISTA (Fast Iterative Shrinkage-Thresholding Algorithm, [54]) to solve the LASSO problem and record the intermediate iteration results. We consider the following experimental setup:

- $n = 300$, $p \in \{1000, 2000, 3000\}$, $\sigma = 1$ and $s^* = 20$.
- Design matrix: $X \sim \mathcal{N}(\mathbf{0}_p, \mathbf{\Sigma})$ with $\mathbf{\Sigma}_{ij} = 0.5^{|i-j|}$.
- True coefficients: $\boldsymbol{\beta}^* \sim \text{Unif}([-3, -1] \cup [1, 3])$.
- Regularization parameters:

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TABLE IV
THE COMPUTATIONAL RESULTS WITH 100 REPETITIONS

n	Method	EE	SE	TPR	FPR($\times$ 100)	MCC
500	FAHTP	**1.815(0.484)**	-0.600(1.325)	0.971 (0.039)	**0.009(0.018)**	**0.980(0.022)**
	ABESS-SIC	1.863(0.454)	-0.200 (1.356)	0.975(0.035)	0.019 (0.024)	0.978(0.020)
	ABESS-IC	1.890(0.461)	0.120 (1.547)	**0.977(0.034)**	0.027 (0.035)	0.975(0.023)
	CD-IC	1.888(0.442)	**0.040(1.525)**	**0.977(0.033)**	0.025(0.032)	0.976(0.021)
	CDPSI-IC	1.886(0.443)	0.050(1.499)	0.977(0.033)	0.023(0.031)	0.976(0.020)
$1,000$	FAHTP	**1.030(0.198)**	**0.120(0.480)**	0.998 (0.008)	**0.006(0.013)**	**0.9996(0.007)**
	ABESS-SIC	1.063(0.198)	0.320(0.653)	**0.999(0.007)**	0.012(0.020)	0.993(0.010)
	ABESS-IC	1.121(0.218)	0.620(0.945)	**0.999(0.007)**	0.022(0.030)	0.989(0.014)
	CD-IC	1.127(0.226)	0.660(1.002)	**0.999(0.007)**	0.024(0.032)	0.988(0.015)
	CDPSI-IC	1.123(0.223)	0.620 (0.991)	**0.999(0.031)**	0.020(0.032)	0.989 (0.017)

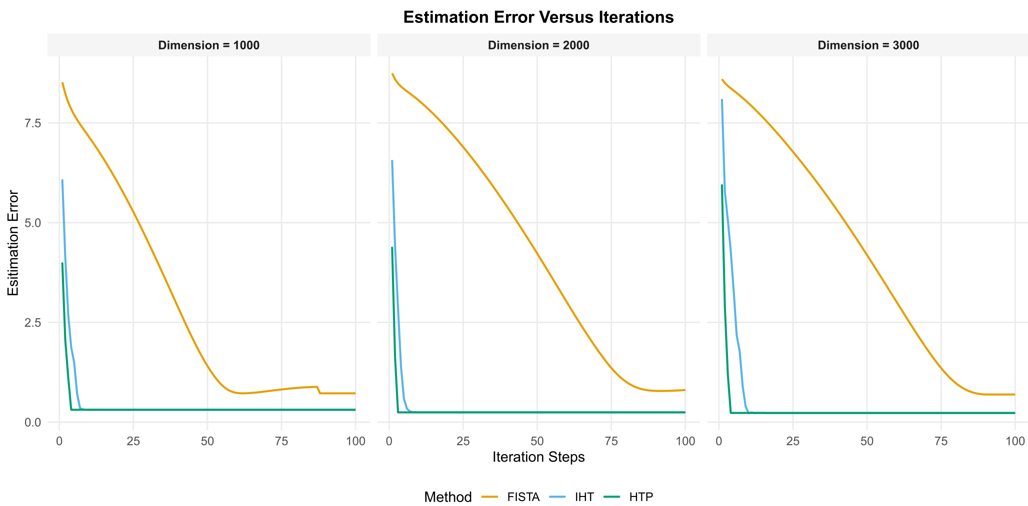

Fig. 16. Convergence behavior of estimation error versus iterations for FISTA, IHT, and HTP.

- LASSO: $\lambda = 100\sigma\sqrt{\log(p/s^*)/n}$.
- IHT/HTP: $s = 20$.

Figure 16 shows that HTP demonstrates the fastest convergence (within 5 iterations), and IHT shows slightly slower convergence than HTP. In comparison, FISTA requires $\approx$ 90 iterations to converge while exhibiting higher estimation error throughout the optimization process.

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